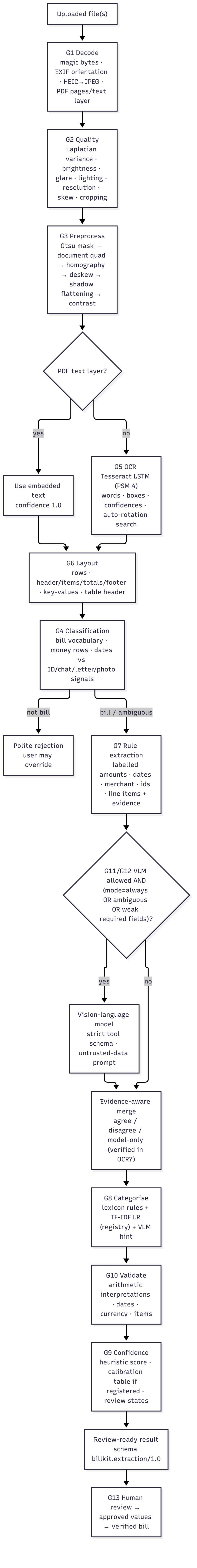
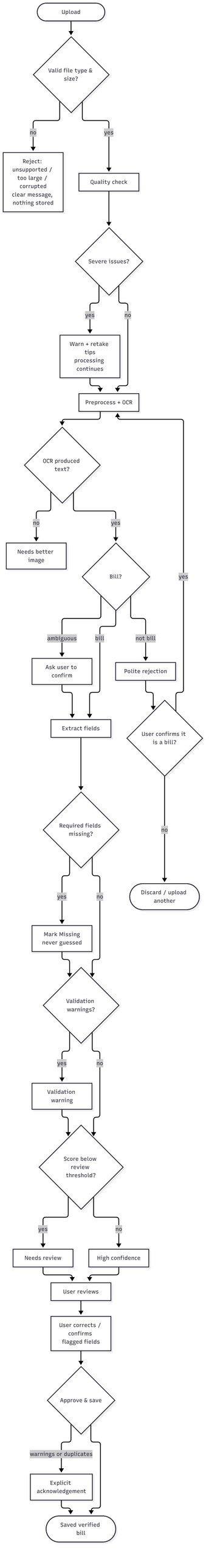
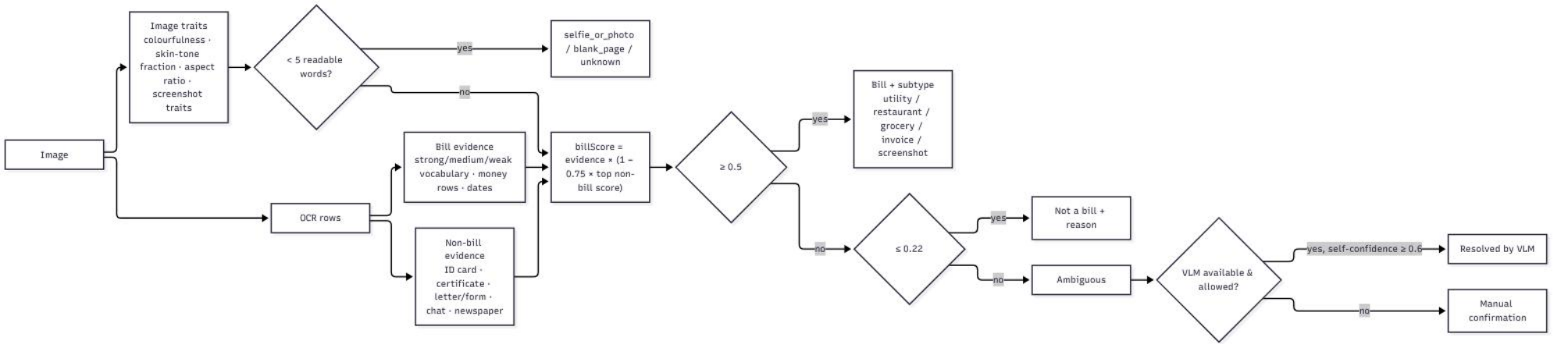
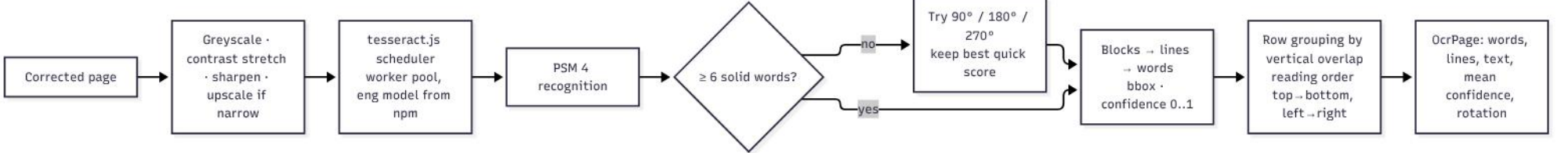
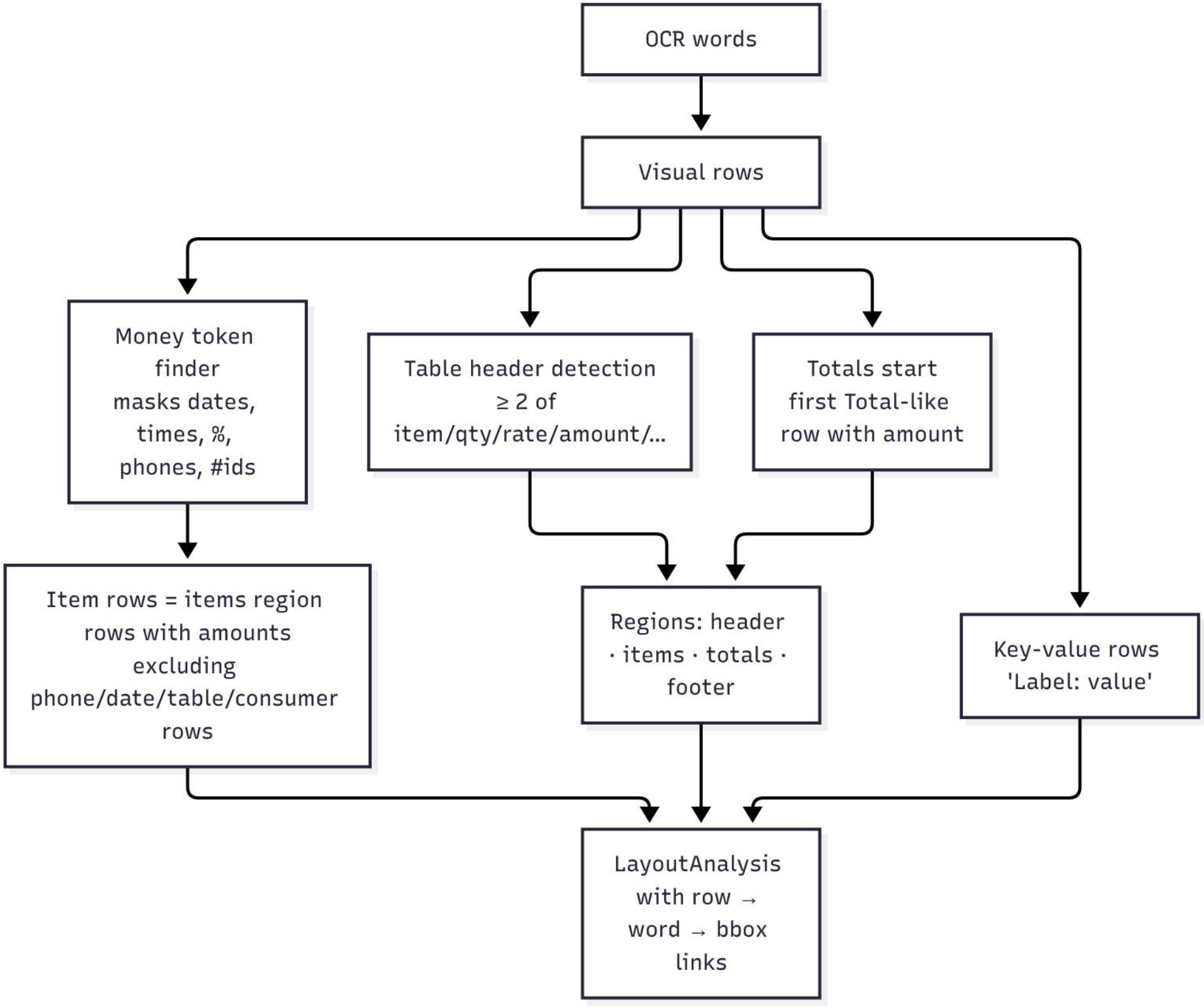

Browser / installed PWA (React + TypeScript)
Review & correct
Camera / gallery / drag-drop
Upload page
Bill history / analytics
IndexedDB offline copies
approved values
JSON result
multipart upload
REST API /api/* auth · CSRF · rate limits
Processing queue
Image storage local disk, UUID paths
BillKit server (Node.js · Fastify)
AI pipeline (@billkit/ai)
Quality assessment
Preprocessing
Tesseract OCR
Bill classifier evidence rules
Extraction rules
Categoriser rules + TF-IDF model
Validation
Confidence
only if configured & allowed
Anthropic vision model optional, opt-in
Operational metrics (no bill content)
SQLite users · bills · corrections · job_metrics

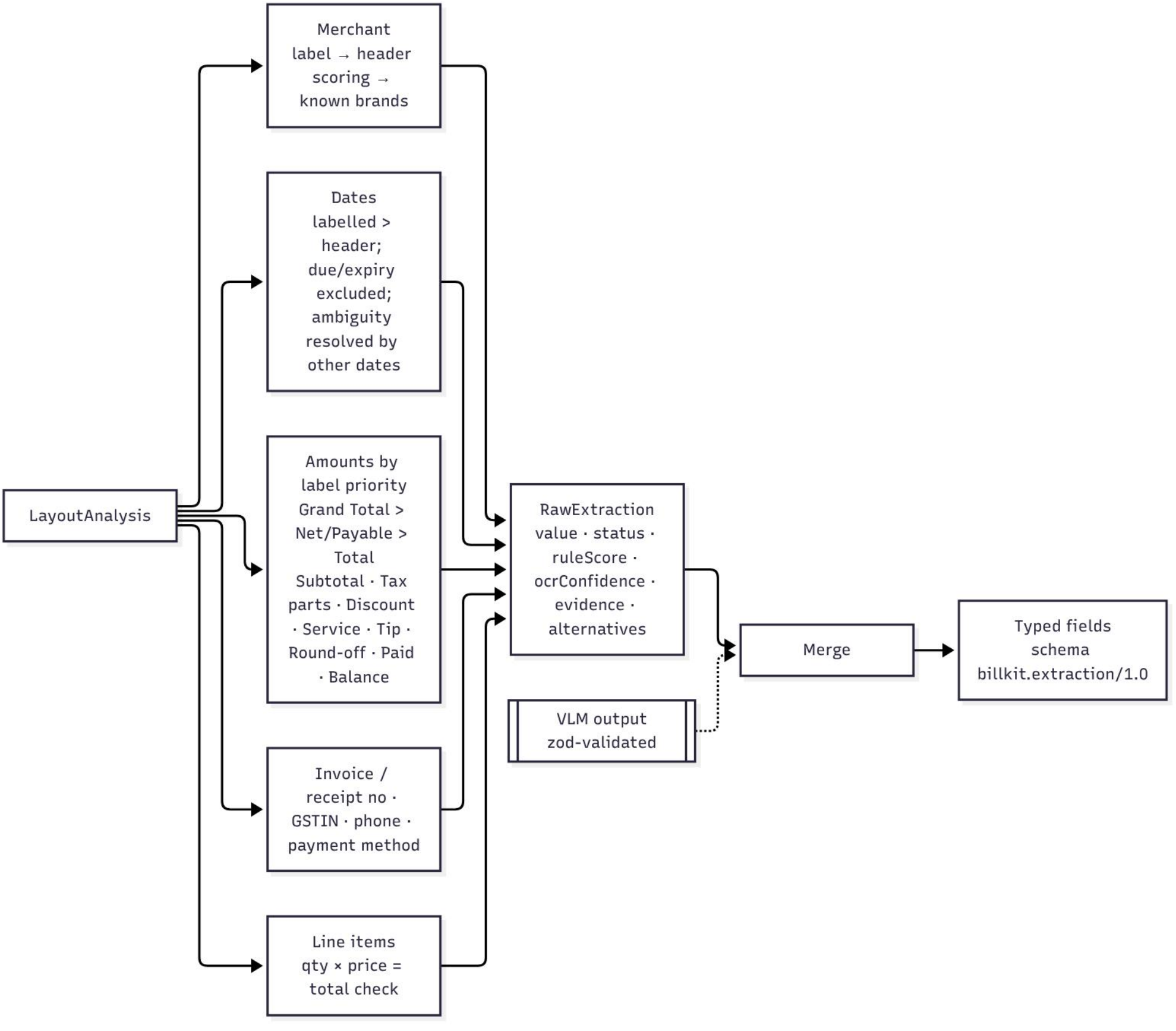

Uploaded file(s)
G1 Decode magic bytes · EXIF orientation · HEIC → JPEG · PDF pages/text layer
G2 Quality Laplacian variance · brightness · glare · lighting · resolution · skew · cropping
G3 Preprocess Otsu mask → document quad → homography → deskew → shadow flattening → contrast
PDF text layer?
yes
no
Use embedded text confidence 1.0
G5 OCR Tesseract LSTM (PSM 4) words · boxes · confidences · auto-rotation search
G6 Layout rows · header/items/totals/footer · key-values · table header
G4 Classification bill vocabulary · money rows · dates vs ID/chat/letter/photo signals
not bill
bill / ambiguous
Polite rejection user may override
G7 Rule extraction labelled amounts · dates · merchant · ids · line items + evidence
G11/G12 VLM allowed AND (mode=always OR ambiguous OR weak required fields)?
yes
no
Vision-language model strict tool schema · untrusted-data prompt
Evidence-aware merge agree / disagree / model-only (verified in OCR?)
G8 Categorise lexicon rules + TF-IDF LR (registry) + VLM hint
G10 Validate arithmetic interpretations · dates · currency · items
G9 Confidence heuristic score · calibration table if registered · review states
Review-ready result schema billkit.extraction/1.0
G13 Human review → approved values → verified bill

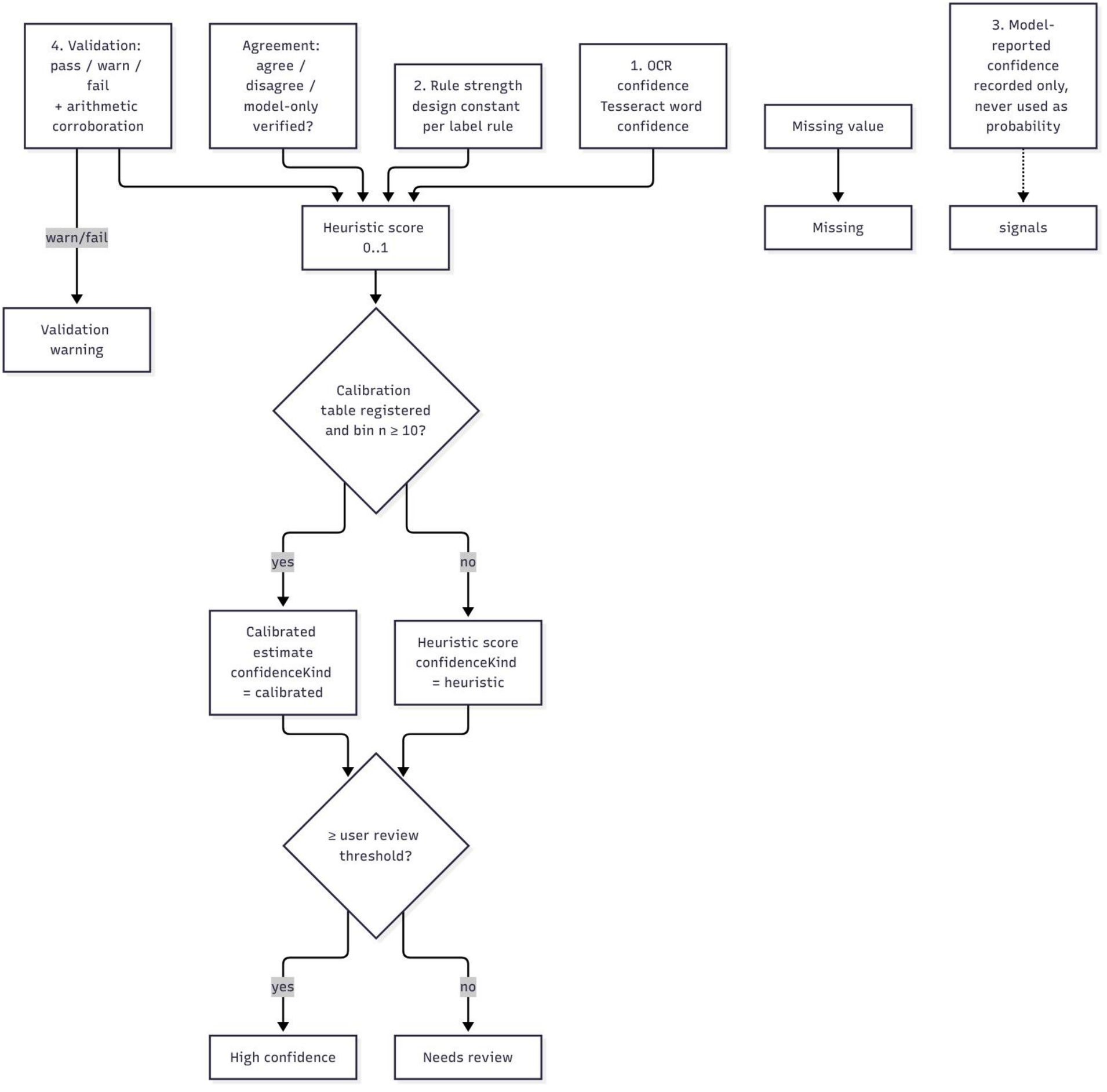

Upload
Valid file type & size?
no
yes
Reject: unsupported / too large / corrupted clear message, nothing stored
Quality check
Severe issues?
yes
no
Warn + retake tips processing continues
Preprocess + OCR
OCR produced text?
no
yes
Needs better image
Bill?
yes
ambiguous
bill
not bill
Ask user to confirm
Polite rejection
Extract fields
User confirms it is a bill?
Required fields missing?
no
yes
no
Discard / upload another
yes
no
Mark Missing never guessed
Validation warnings?
yes
no
Validation warning
Score below review threshold?
yes
no
Needs review
High confidence
User reviews
User corrects / confirms flagged fields
Approve & save
warnings or duplicates
Explicit acknowledgement
Saved verified bill

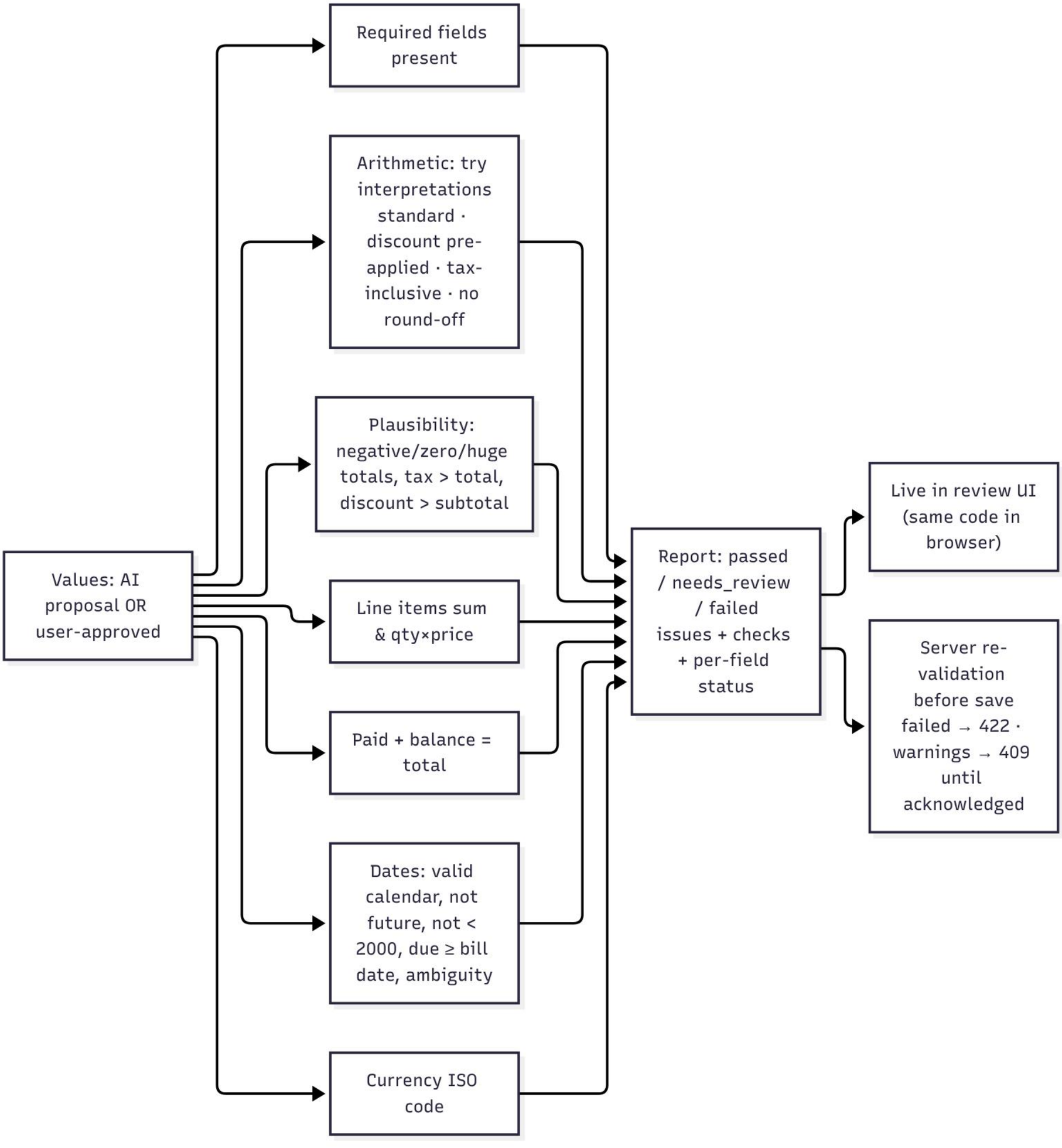

Image
Image traits
colourfulness · skin-tone fraction · aspect ratio · screenshot traits
< 5 readable words?
yes
selfie_or_photo / blank_page / unknown
no
OCR rows
Bill evidence
strong/medium/weak vocabulary · money rows · dates
Non-bill evidence
ID card · certificate · letter/form · chat · newspaper
billScore = evidence × (1 − 0.75 × top non-bill score)
≥ 0.5
yes
Bill + subtype
utility / restaurant / grocery / invoice / screenshot
no
≤ 0.22
yes
Not a bill + reason
no
Ambiguous
VLM available & allowed?
yes, self-confidence ≥ 0.6
Resolved by VLM
no
Manual confirmation

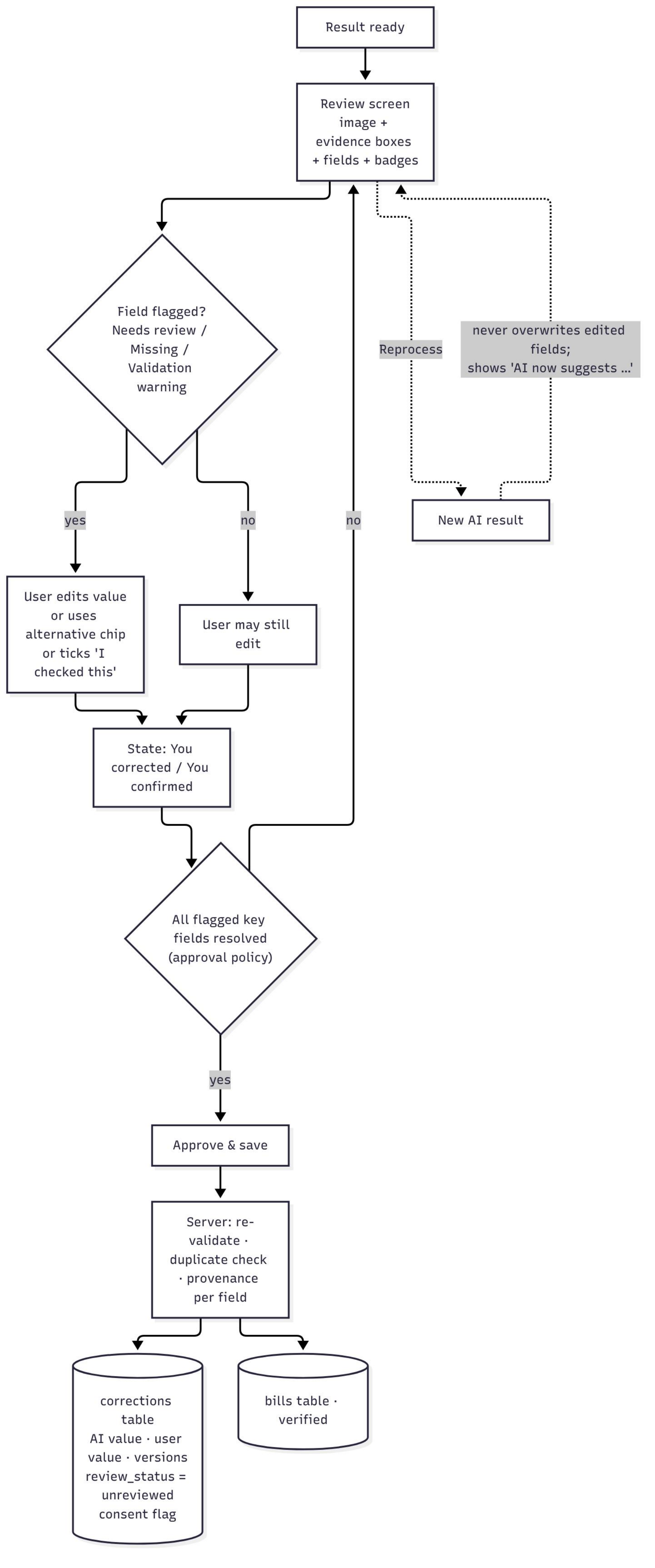

Corrected page
Greyscale · contrast stretch · sharpen · upscale if narrow
tesseract.js scheduler worker pool, eng model from npm
PSM 4 recognition
≥ 6 solid words?
no
yes
Try 90° / 180° / 270° keep best quick score
Blocks → lines → words bbox · confidence 0..1
Row grouping by vertical overlap reading order top→bottom, left→right
OcrPage: words, lines, text, mean confidence, rotation

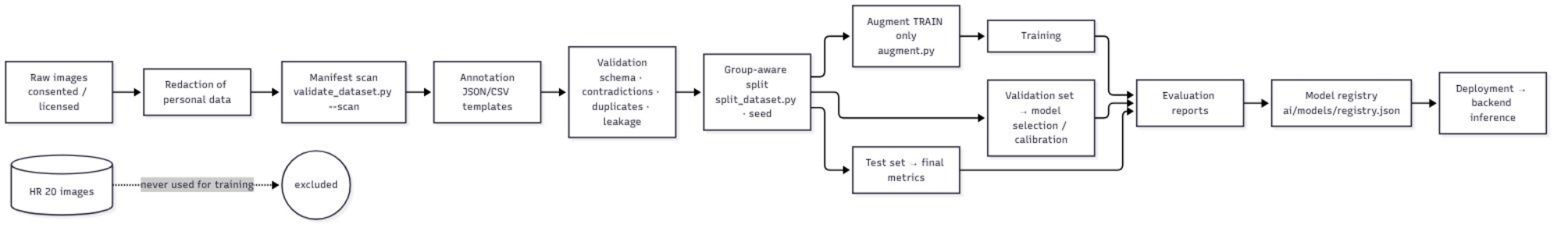

OCR words
Visual rows
Money token finder masks dates, times, %, phones, #ids
Table header detection ≥ 2 of item/qty/rate/amount/...
Totals start first Total-like row with amount
Item rows = items region rows with amounts excluding phone/date/table/consumer rows
Regions: header · items · totals · footer
Key-value rows 'Label: value'
LayoutAnalysis with row → word → bbox links

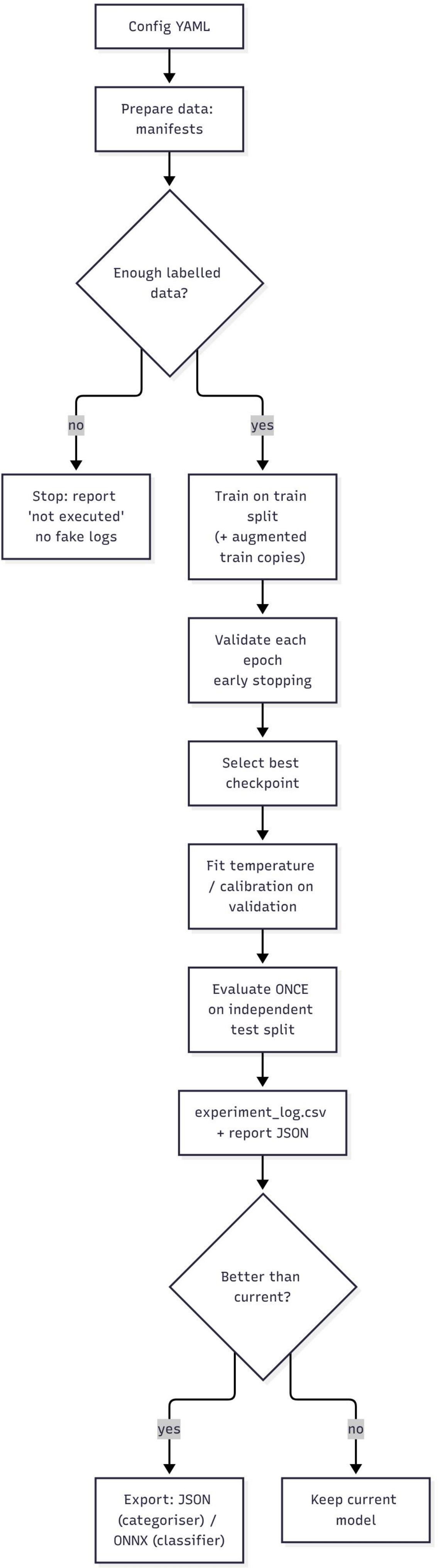

LayoutAnalysis
Merchant
label → header
scoring →
known brands
Dates
labelled >
header;
due/expiry
excluded;
ambiguity
resolved by
other dates
Amounts by
label priority
Grand Total >
Net/Payable >
Total
Subtotal · Tax
parts · Discount
· Service · Tip ·
Round-off · Paid
· Balance
Invoice /
receipt no ·
GSTIN · phone ·
payment method
Line items
qty × price =
total check
RawExtraction
value · status ·
ruleScore ·
ocrConfidence ·
evidence ·
alternatives
VLM output
zod-validated
Merge
Typed fields
schema
billkit.extraction/1.0

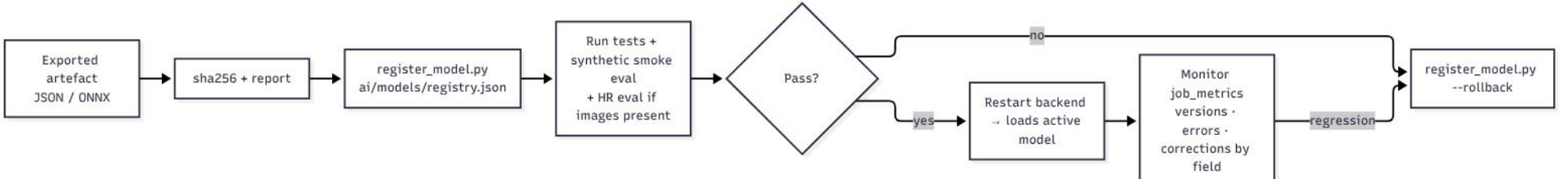

4. Validation: pass / warn / fail + arithmetic corroboration
Agreement: agree / disagree / model-only verified?
2. Rule strength design constant per label rule
1. OCR confidence Tesseract word confidence
Missing value
3. Model-reported confidence recorded only, never used as probability
warn/fail
Heuristic score 0..1
Missing
signals
Validation warning
Calibration table registered and bin n ≥ 10?
yes
no
Calibrated estimate confidenceKind = calibrated
Heuristic score confidenceKind = heuristic
≥ user review threshold?
yes
no
High confidence
Needs review

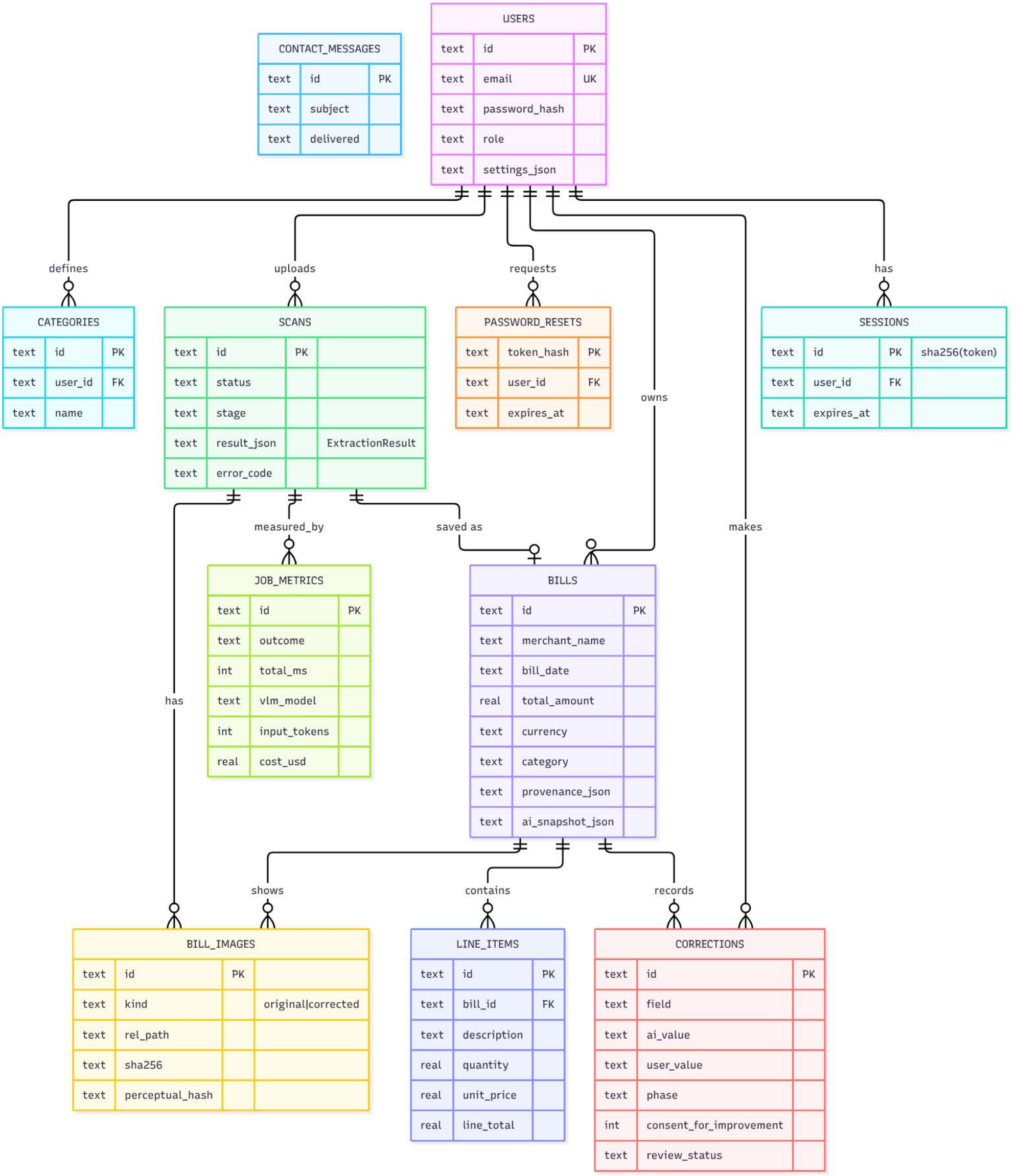

Values: AI proposal OR user-approved
Required fields present
Arithmetic: try interpretations standard · discount pre-applied · tax-inclusive · no round-off
Plausibility: negative/zero/huge totals, tax > total, discount > subtotal
Line items sum & qty×price
Paid + balance = total
Dates: valid calendar, not future, not < 2000, due ≥ bill date, ambiguity
Currency ISO code
Report: passed / needs_review / failed issues + checks + per-field status
Live in review UI (same code in browser)
Server re-validation before save failed → 422 · warnings → 409 until acknowledged

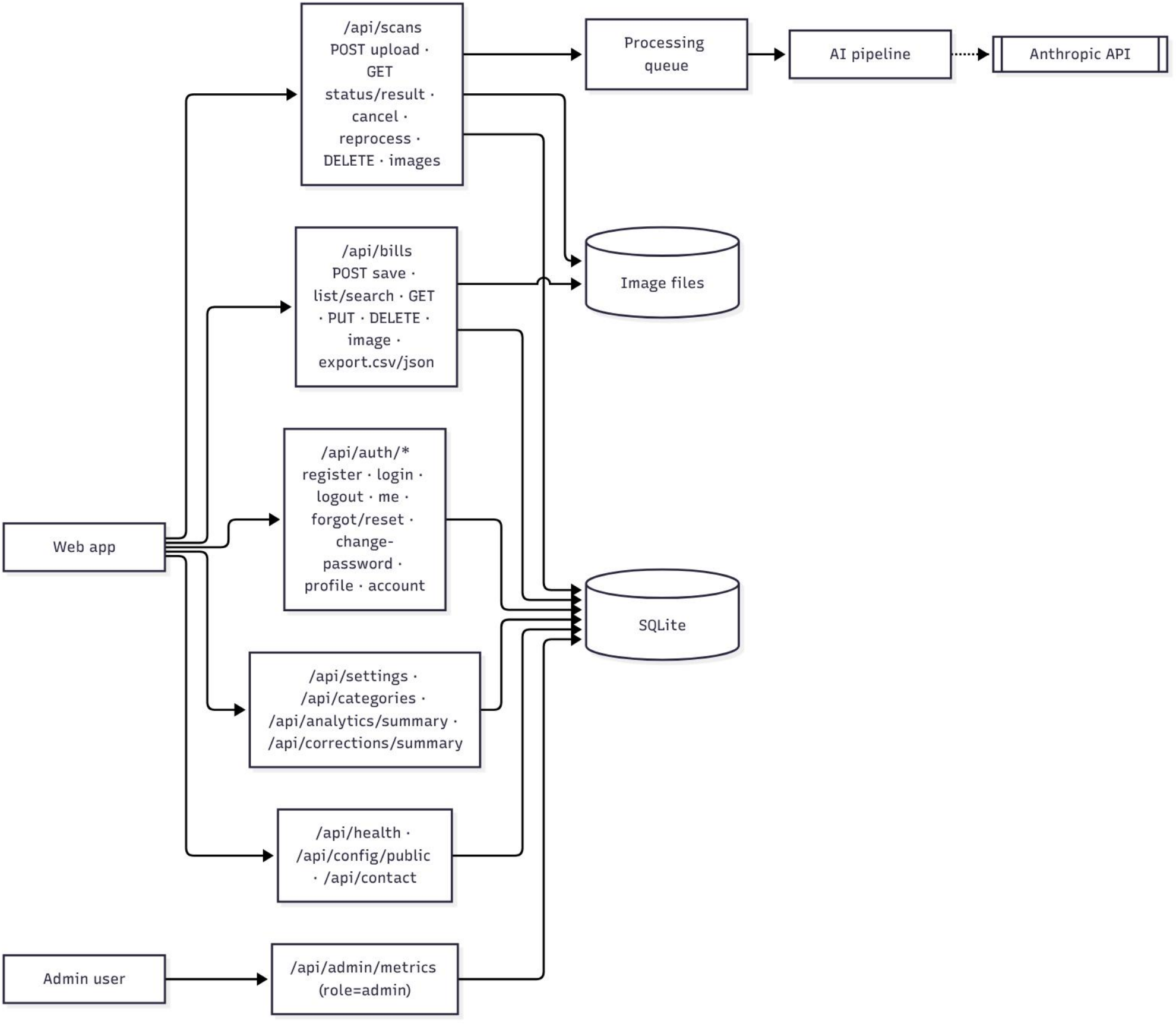

Result ready
Review screen image + evidence boxes + fields + badges
Field flagged? Needs review / Missing / Validation warning
yes
no
no
Reprocess
never overwrites edited fields; shows 'AI now suggests ...'
New AI result
User edits value or uses alternative chip or ticks 'I checked this'
User may still edit
State: You corrected / You confirmed
All flagged key fields resolved (approval policy)
yes
Approve & save
Server: re-validate · duplicate check · provenance per field
corrections table
AI value · user value · versions
review_status = unreviewed
consent flag
bills table · verified

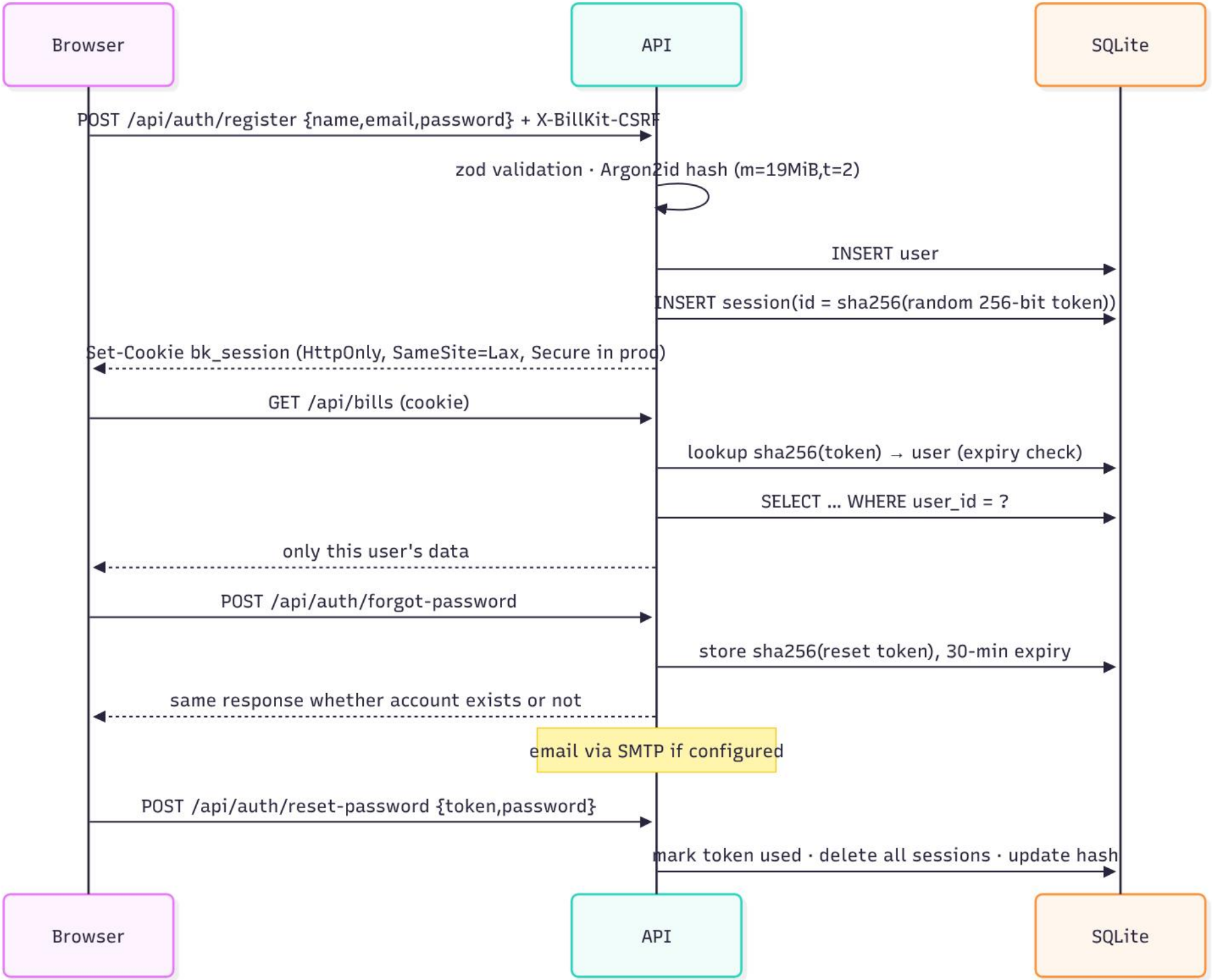

Raw images consented / licensed
Redaction of personal data
Manifest scan validate_dataset.py --scan
Annotation JSON/CSV templates
Validation schema · contradictions · duplicates · leakage
Group-aware split split_dataset.py · seed
Augment TRAIN only augment.py
Training
Validation set → model selection / calibration
Test set → final metrics
Evaluation reports
Model registry ai/models/registry.json
Deployment → backend inference
HR 20 images
never used for training
excluded

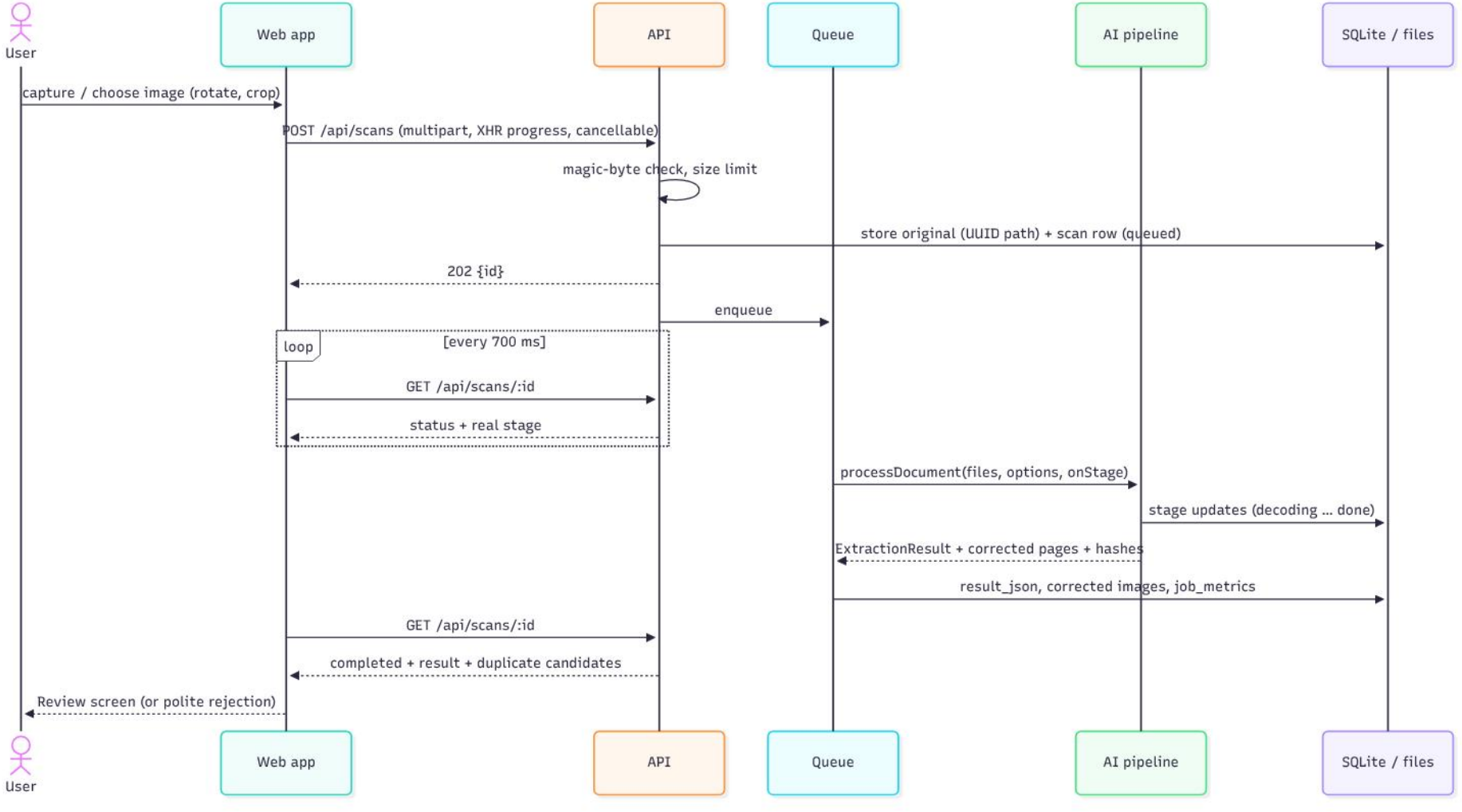

Config YAML
Prepare data: manifests
Enough labelled data?
no
yes
Stop: report 'not executed' no fake logs
Train on train split (+ augmented train copies)
Validate each epoch early stopping
Select best checkpoint
Fit temperature / calibration on validation
Evaluate ONCE on independent test split
experiment_log.csv + report JSON
Better than current?
yes
no
Export: JSON (categoriser) / ONNX (classifier)
Keep current model

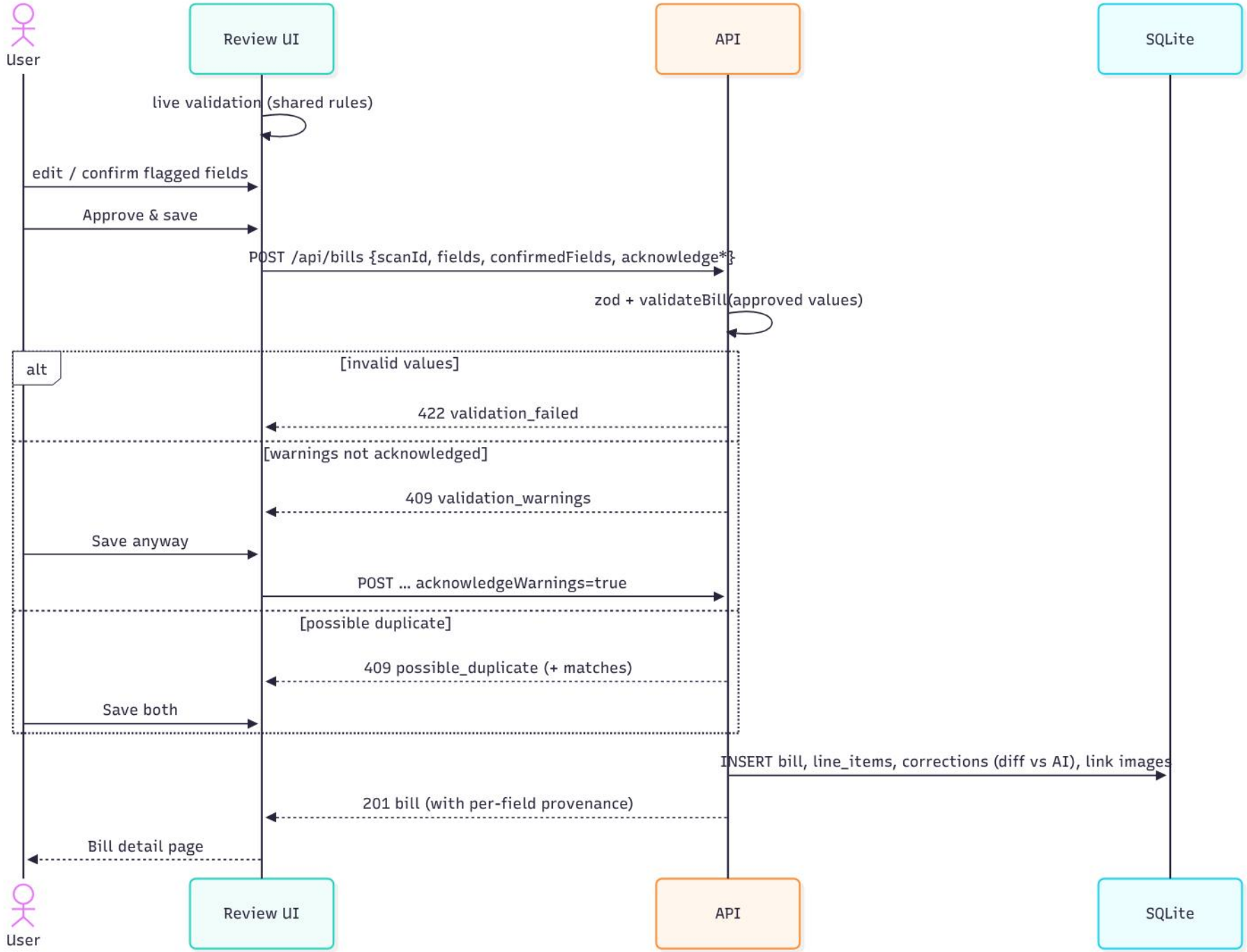

Exported artefact JSON / ONNX
sha256 + report
register_model.py ai/models/registry.json
Run tests + synthetic smoke eval + HR eval if images present
Pass?
no
yes
Restart backend → loads active model
Monitor job_metrics versions · errors · corrections by field
regression
register_model.py --rollback

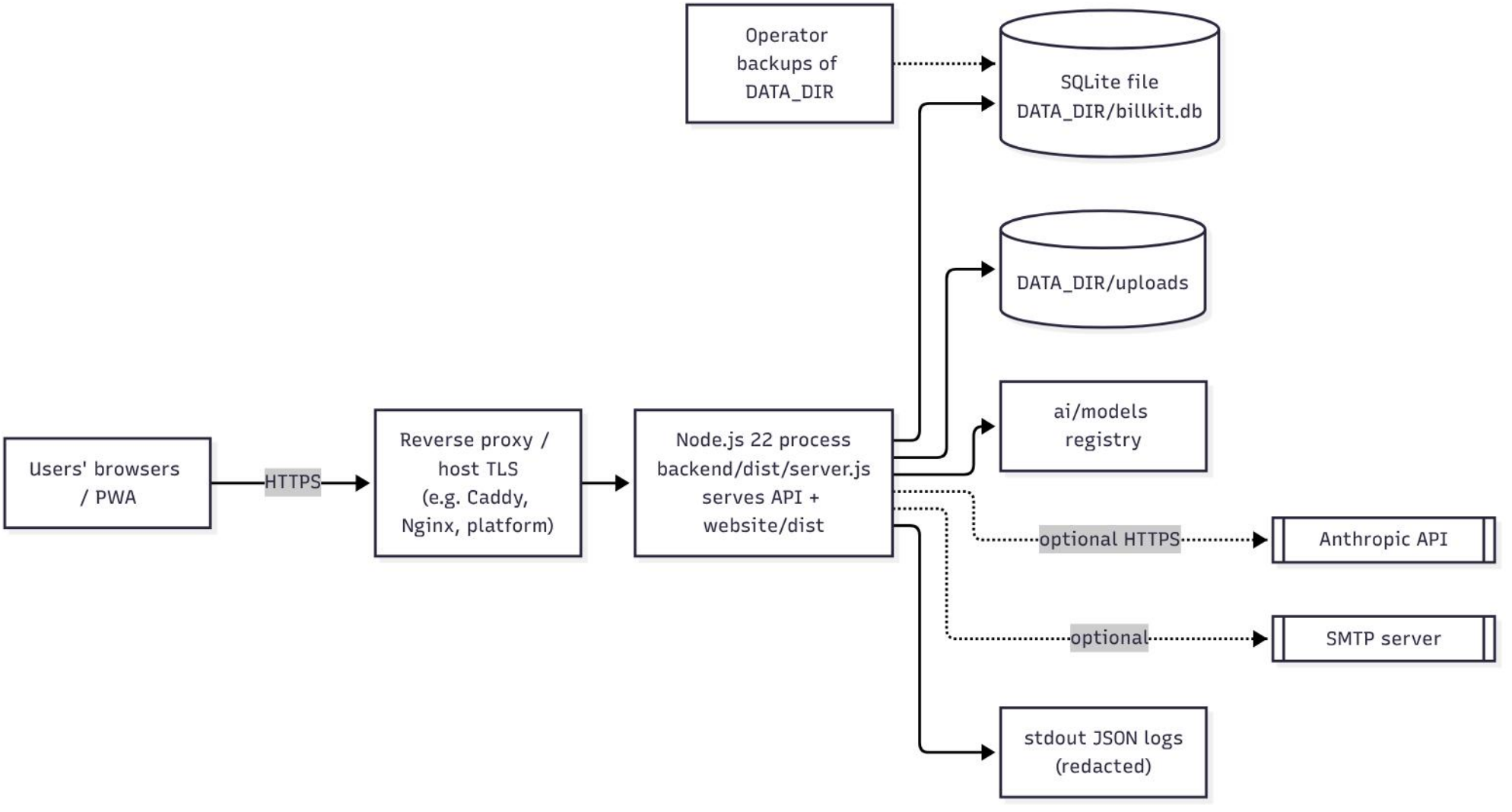

CONTACT_MESSAGES
text id PK
text subject
text delivered
USERS
text id PK
text email UK
text password_hash
text role
text settings_json
defines
uploads
requests
has
owns
makes
CATEGORIES
text id PK
text user_id FK
text name
SCANS
text id PK
text status
text stage
text result_json ExtractionResult
text error_code
PASSWORD_RESETS
text token_hash PK
text user_id FK
text expires_at
SESSIONS
text id PK sha256(token)
text user_id FK
text expires_at
measured_by
saved as
has
JOB_METRICS
text id PK
text outcome
int total_ms
text vlm_model
int input_tokens
real cost_usd
BILLS
text id PK
text merchant_name
text bill_date
real total_amount
text currency
text category
text provenance_json
text ai_snapshot_json
shows
contains
records
BILL_IMAGES
text id PK
text kind original|corrected
text rel_path
text sha256
text perceptual_hash
LINE_ITEMS
text id PK
text bill_id FK
text description
real quantity
real unit_price
real line_total
CORRECTIONS
text id PK
text field
text ai_value
text user_value
text phase
int consent_for_improvement
text review_status

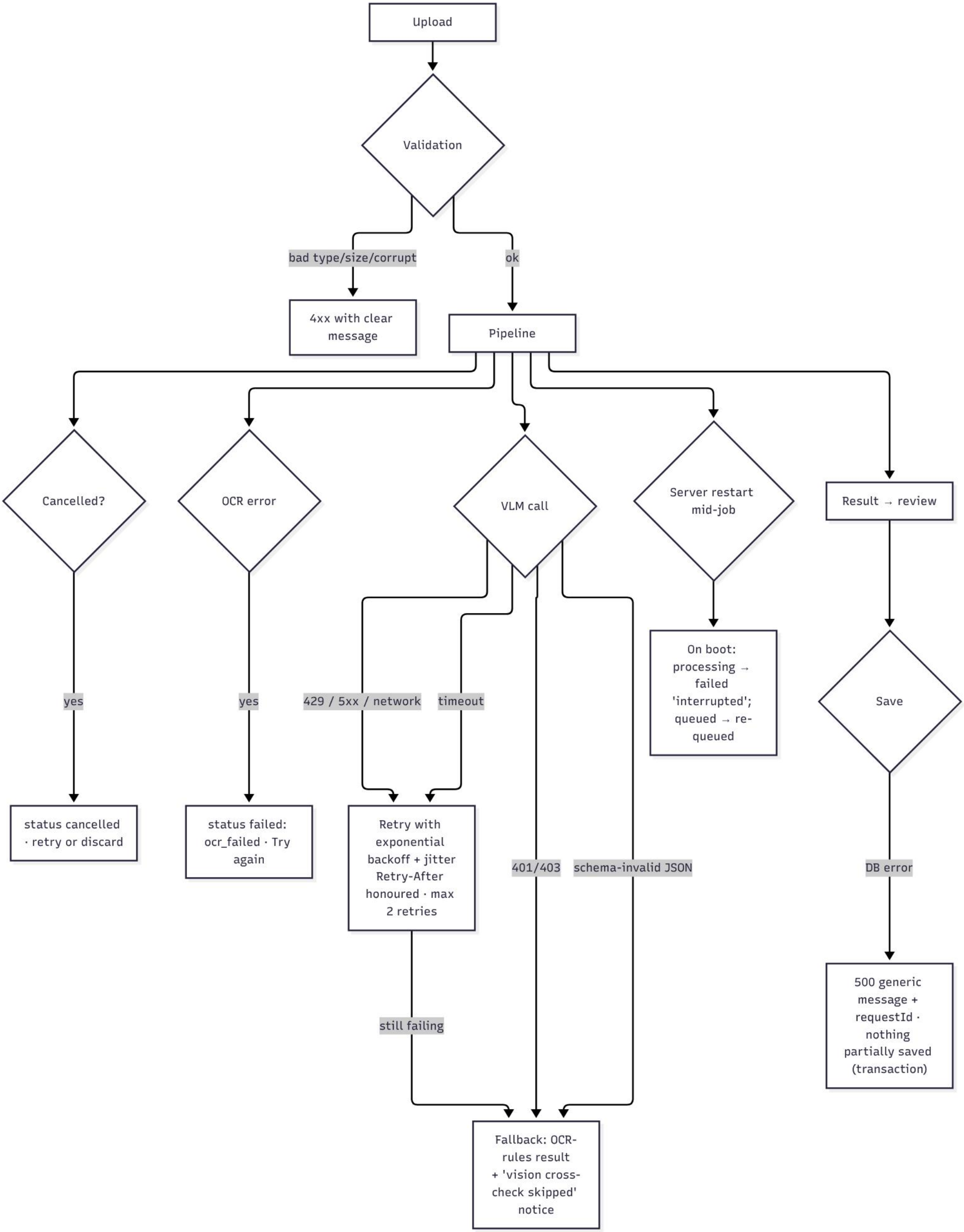

/api/scans
POST upload · GET status/result · cancel · reprocess · DELETE · images
Processing queue
AI pipeline
Anthropic API
/api/bills
POST save · list/search · GET · PUT · DELETE · image · export.csv/json
Image files
/api/auth/*
register · login · logout · me · forgot/reset · change-password · profile · account
Web app
SQLite
/api/settings · /api/categories · /api/analytics/summary · /api/corrections/summary
/api/health · /api/config/public · /api/contact
Admin user
/api/admin/metrics (role=admin)

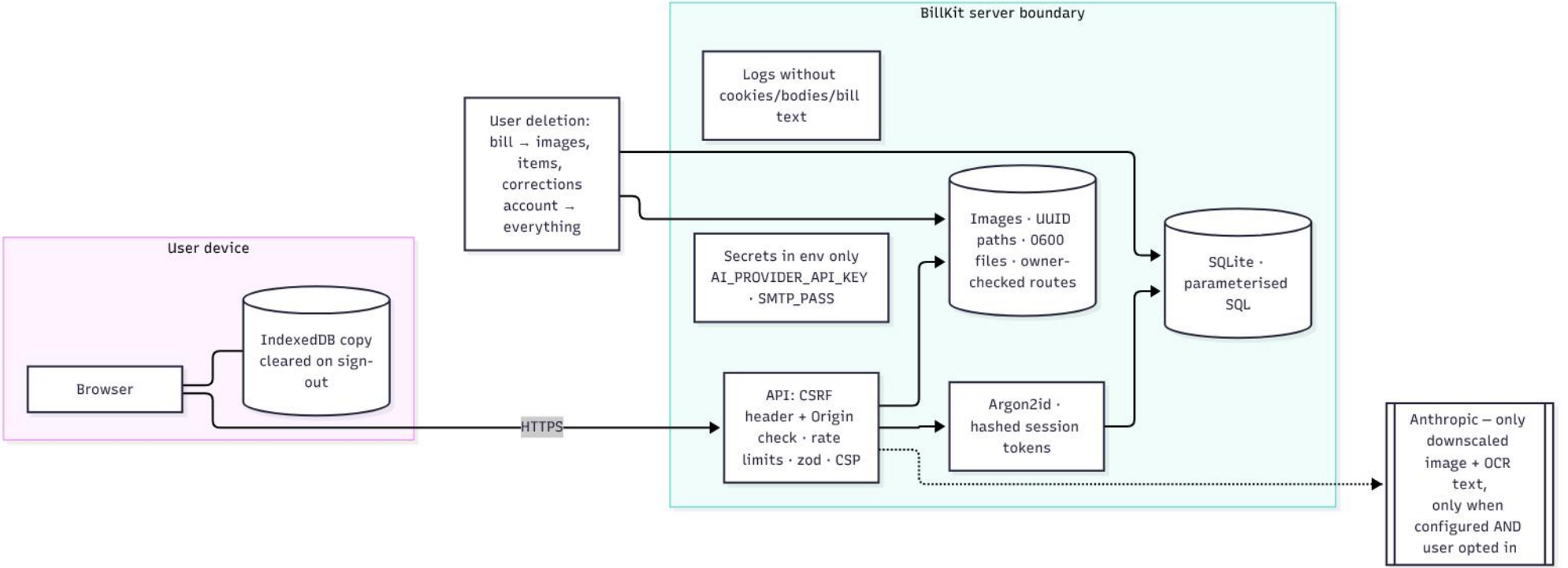

Browser
API
SQLite
POST /api/auth/register {name,email,password} + X-BillKit-CSRF
zod validation · Argon2id hash (m=19MiB,t=2)
INSERT user
INSERT session(id = sha256(random 256-bit token))
Set-Cookie bk_session (HttpOnly, SameSite=Lax, Secure in prod)
GET /api/bills (cookie)
lookup sha256(token) → user (expiry check)
SELECT … WHERE user_id = ?
only this user's data
POST /api/auth/forgot-password
store sha256(reset token), 30-min expiry
same response whether account exists or not
email via SMTP if configured
POST /api/auth/reset-password {token,password}
mark token used · delete all sessions · update hash
Browser
API
SQLite

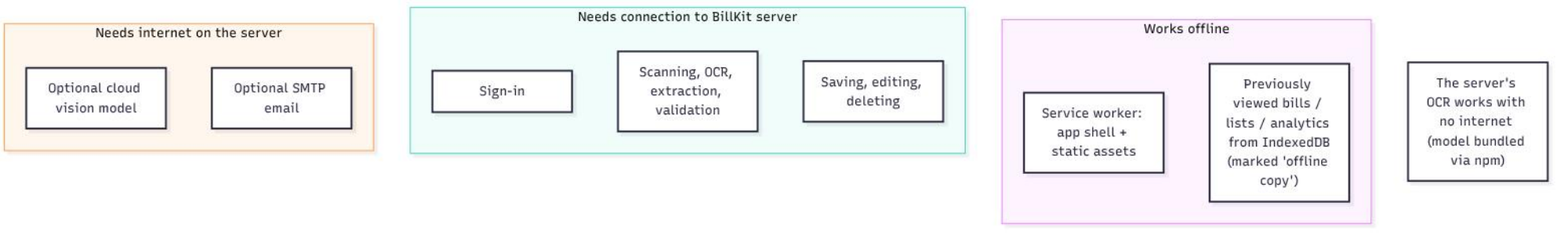

User
Web app
API
Queue
AI pipeline
SQLite / files
capture / choose image (rotate, crop)
POST /api/scans (multipart, XHR progress, cancellable)
magic-byte check, size limit
store original (UUID path) + scan row (queued)
202 {id}
enqueue
loop
[every 700 ms]
GET /api/scans/:id
status + real stage
processDocument(files, options, onStage)
stage updates (decoding … done)
ExtractionResult + corrected pages + hashes
result_json, corrected images, job_metrics
GET /api/scans/:id
completed + result + duplicate candidates
Review screen (or polite rejection)
User
Web app
API
Queue
AI pipeline
SQLite / files

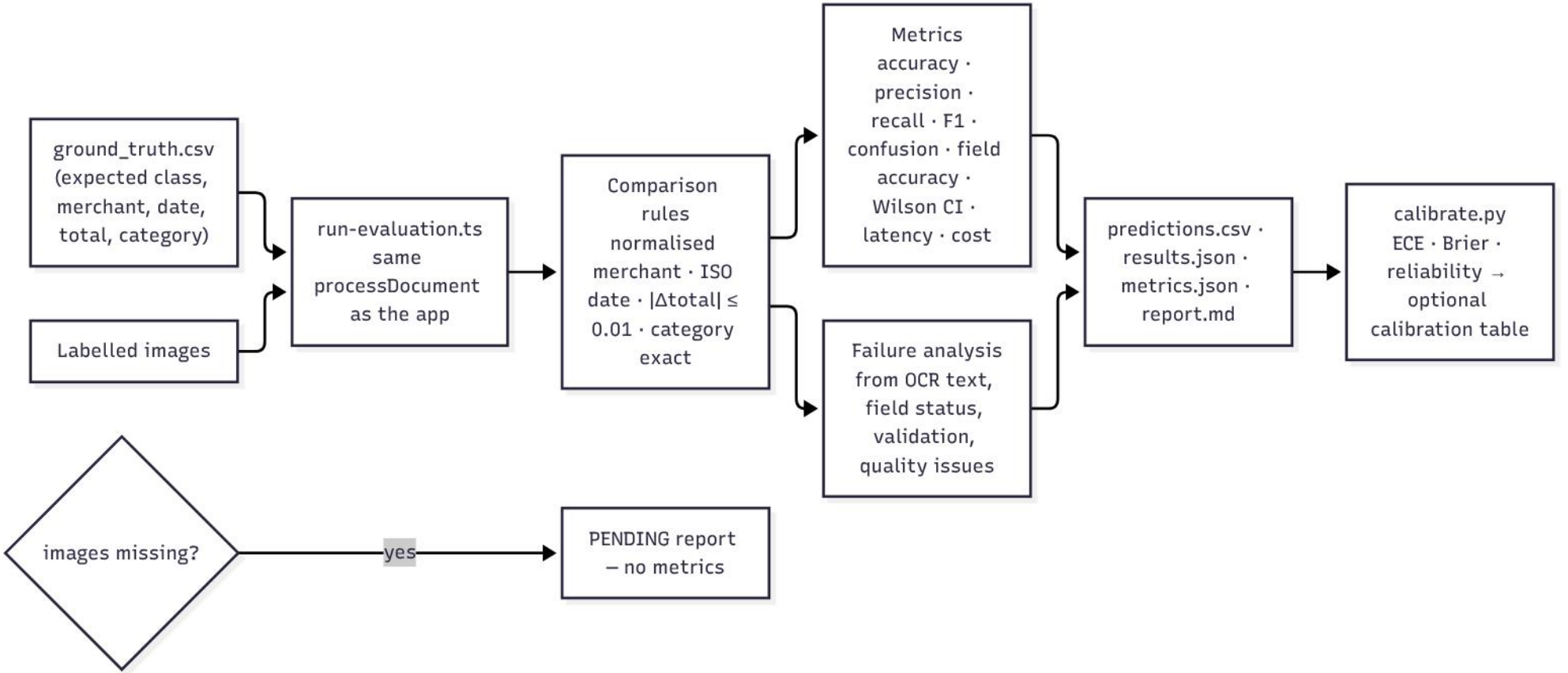

User
Review UI
API
SQLite
live validation (shared rules)
edit / confirm flagged fields
Approve & save
POST /api/bills {scanId, fields, confirmedFields, acknowledge*}
zod + validateBill(approved values)
alt
[invalid values]
422 validation_failed
[warnings not acknowledged]
409 validation_warnings
Save anyway
POST … acknowledgeWarnings=true
[possible duplicate]
409 possible_duplicate (+ matches)
Save both
INSERT bill, line_items, corrections (diff vs AI), link images
201 bill (with per-field provenance)
Bill detail page
User
Review UI
API
SQLite

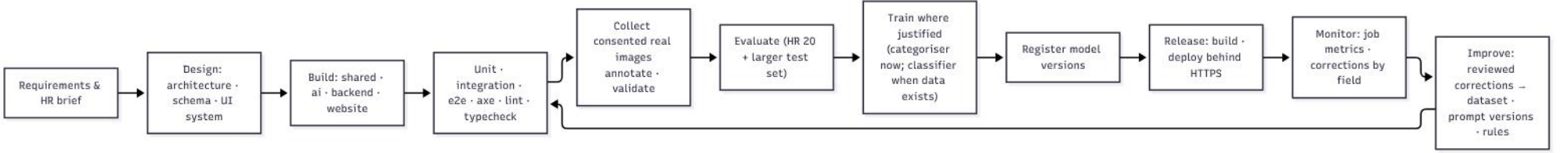

Operator backups of DATA_DIR
SQLite file DATA_DIR/billkit.db
DATA_DIR/uploads
ai/models registry
Users' browsers / PWA
HTTPS
Reverse proxy / host TLS (e.g. Caddy, Nginx, platform)
Node.js 22 process backend/dist/server.js serves API + website/dist
optional HTTPS
Anthropic API
optional
SMTP server
stdout JSON logs (redacted)

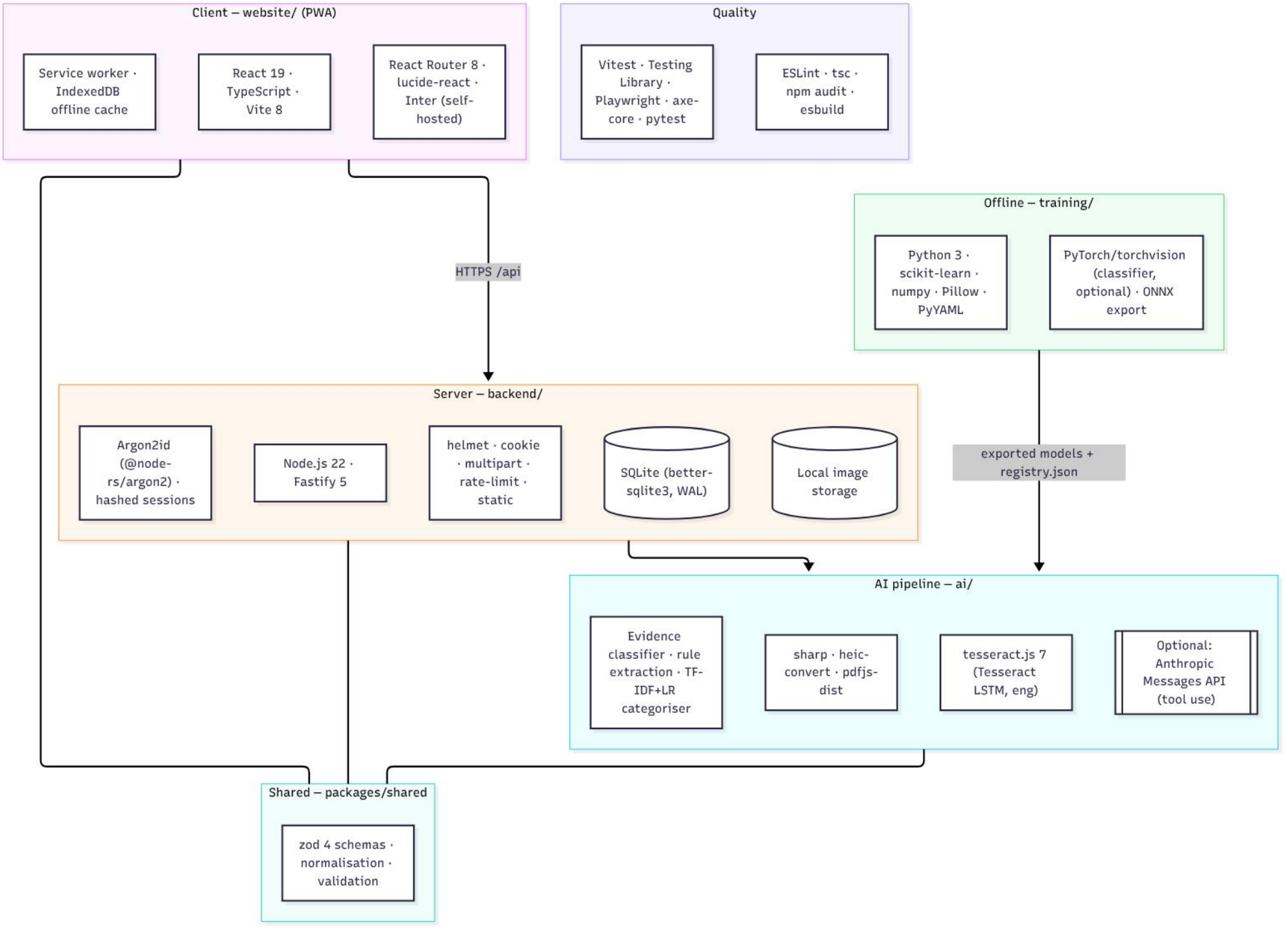

Upload
Validation
bad type/size/corrupt
ok
4xx with clear message
Pipeline
Cancelled?
OCR error
VLM call
Server restart mid-job
Result → review
yes
yes
429 / 5xx / network
timeout
401/403
schema-invalid JSON
On boot: processing → failed 'interrupted'; queued → re-queued
Save
status cancelled · retry or discard
status failed: ocr_failed · Try again
Retry with exponential backoff + jitter Retry-After honoured · max 2 retries
still failing
DB error
Fallback: OCR-rules result + 'vision cross-check skipped' notice
500 generic message + requestId · nothing partially saved (transaction)

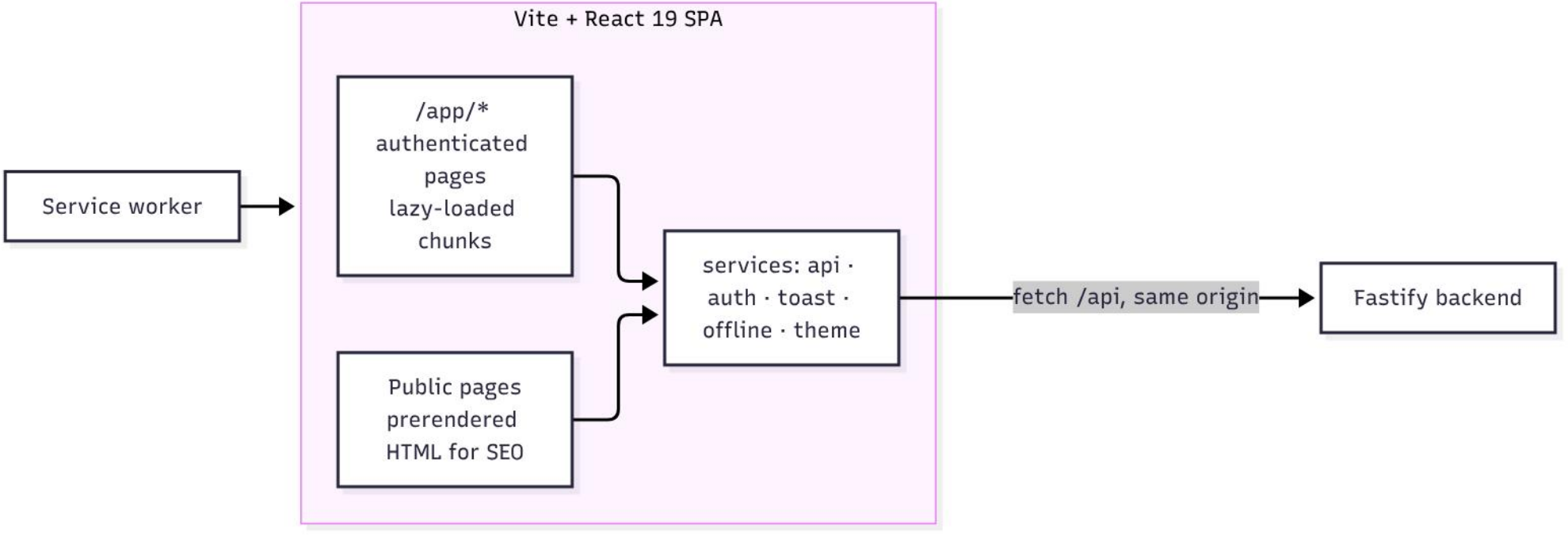

BillKit server boundary
Logs without cookies/bodies/bill text
User deletion: bill → images, items, corrections account → everything
Images · UUID paths · 0600 files · owner-checked routes
SQLite · parameterised SQL
Secrets in env only AI_PROVIDER_API_KEY · SMTP_PASS
User device
IndexedDB copy cleared on sign-out
Browser
HTTPS
API: CSRF header + Origin check · rate limits · zod · CSP
Argon2id · hashed session tokens
Anthropic — only downscaled image + OCR text, only when configured AND user opted in

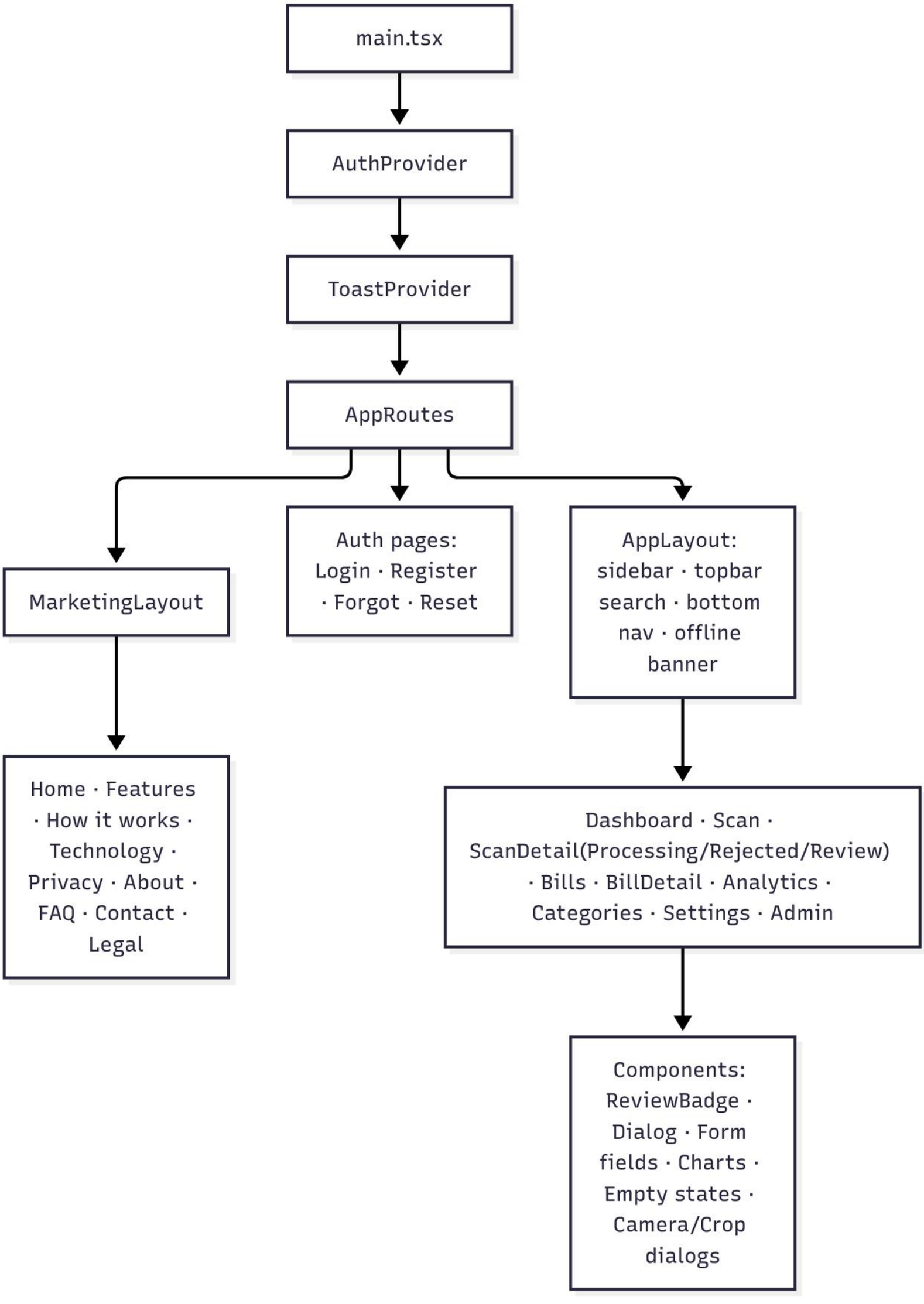

Needs internet on the server
Optional cloud vision model
Optional SMTP email
Needs connection to BillKit server
Sign-in
Scanning, OCR, extraction, validation
Saving, editing, deleting
Works offline
Service worker: app shell + static assets
Previously viewed bills / lists / analytics from IndexedDB (marked 'offline copy')
The server's OCR works with no internet (model bundled via npm)

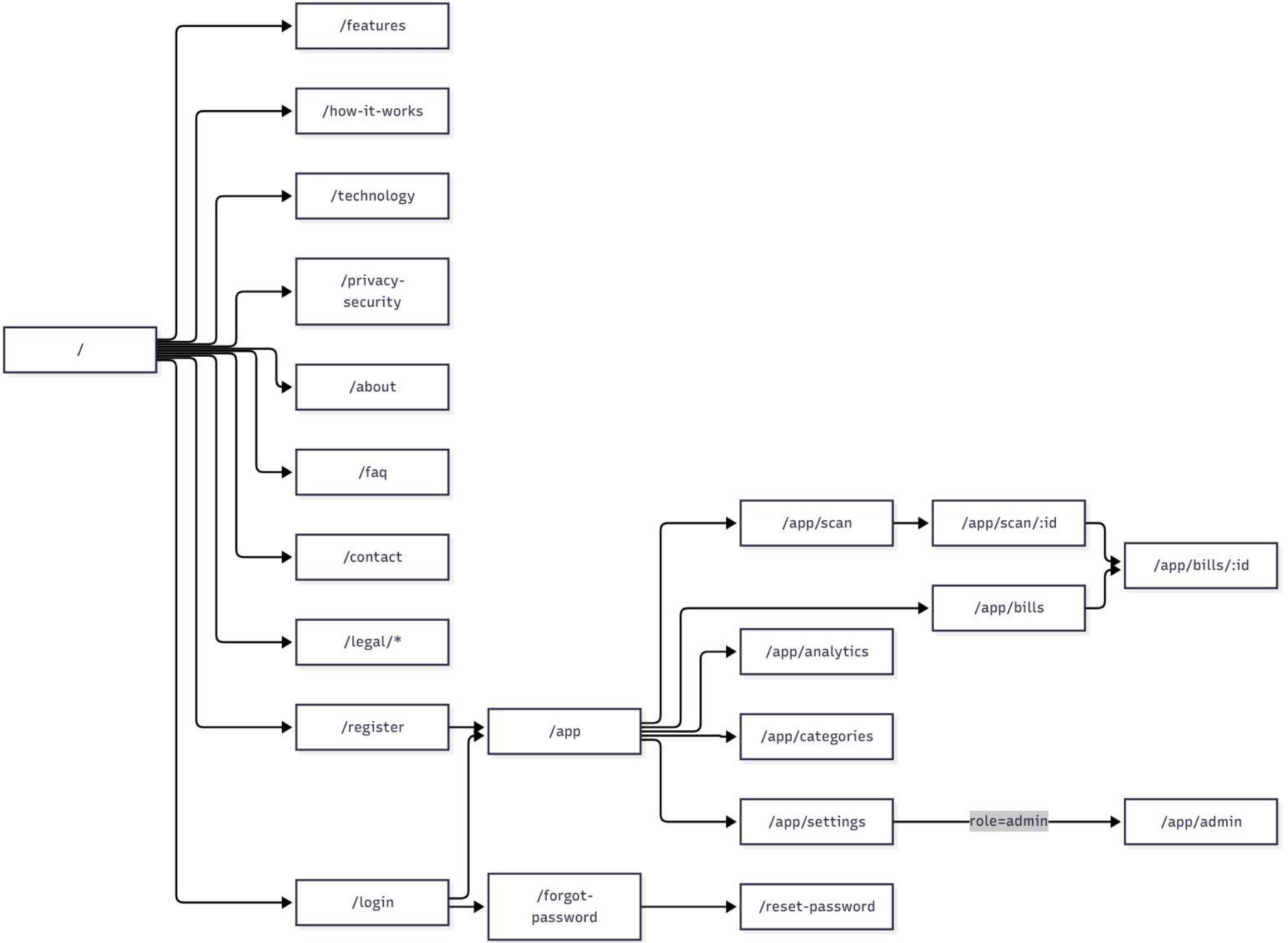

ground_truth.csv (expected class, merchant, date, total, category)
Labelled images
run-evaluation.ts same processDocument as the app
Comparison rules normalised merchant · ISO date · |Δtotal| ≤ 0.01 · category exact
Metrics accuracy · precision · recall · F1 · confusion · field accuracy · Wilson CI · latency · cost
Failure analysis from OCR text, field status, validation, quality issues
predictions.csv · results.json · metrics.json · report.md
calibrate.py ECE · Brier · reliability → optional calibration table
images missing?
yes
PENDING report — no metrics

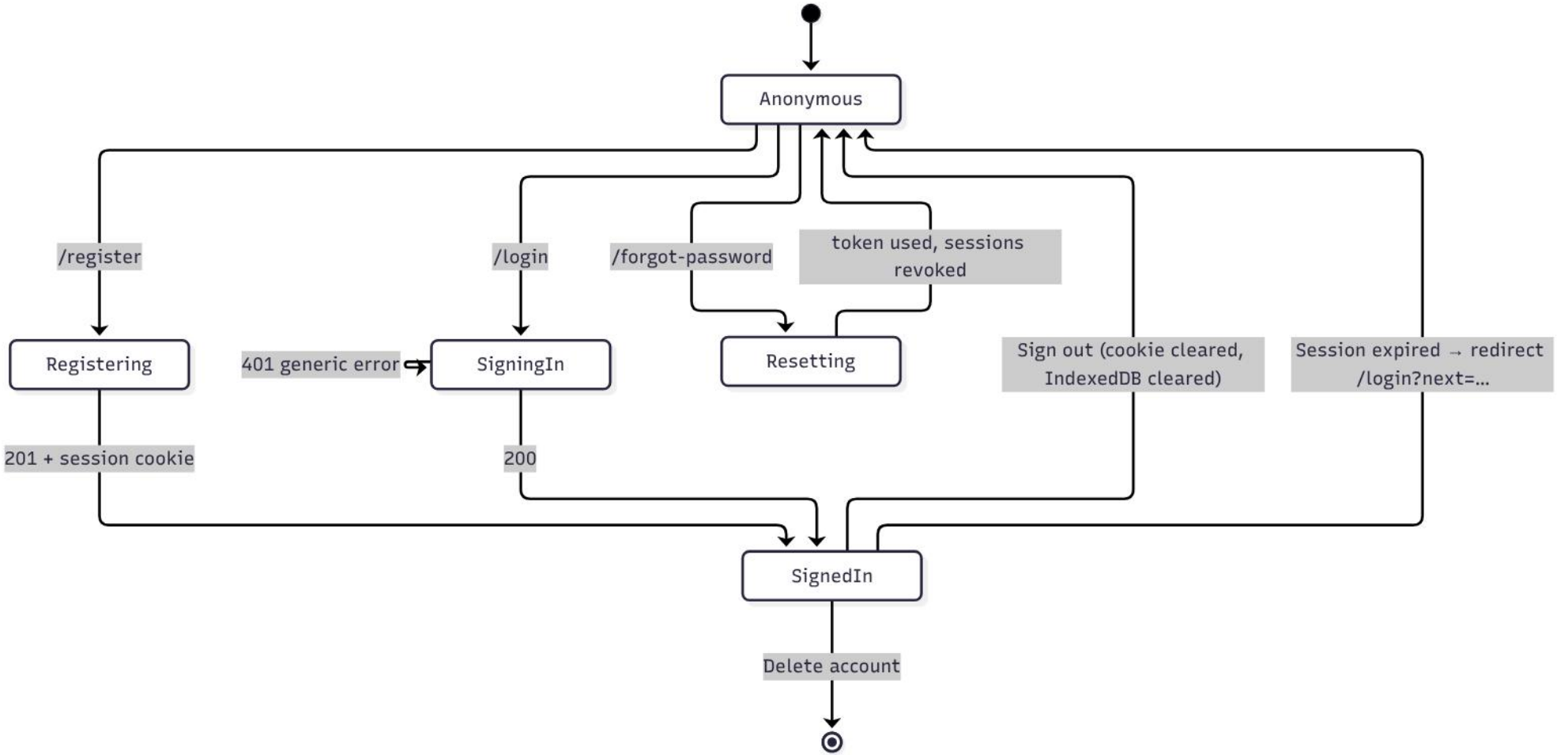

Requirements & HR brief
Design: architecture · schema · UI system
Build: shared · ai · backend · website
Unit · integration · e2e · axe · lint · typecheck
Collect consented real images annotate · validate
Evaluate (HR 20 + larger test set)
Train where justified (categoriser now; classifier when data exists)
Register model versions
Release: build · deploy behind HTTPS
Monitor: job metrics · corrections by field
Improve: reviewed corrections → dataset · prompt versions · rules


Client — website/ (PWA)
Service worker · IndexedDB offline cache
React 19 · TypeScript · Vite 8
React Router 8 · lucide-react · Inter (self-hosted)
Quality
Vitest · Testing Library · Playwright · axe-core · pytest
ESLint · tsc · npm audit · esbuild
Offline — training/
Python 3 · scikit-learn · numpy · Pillow · PyYAML
PyTorch/torchvision (classifier, optional) · ONNX export
HTTPS /api
Server — backend/
Argon2id (@node-rs/argon2) · hashed sessions
Node.js 22 · Fastify 5
helmet · cookie · multipart · rate-limit · static
SQLite (better-sqlite3, WAL)
Local image storage
exported models + registry.json
AI pipeline — ai/
Evidence classifier · rule extraction · TF-IDF+LR categoriser
sharp · heic-convert · pdfjs-dist
tesseract.js 7 (Tesseract LSTM, eng)
Optional: Anthropic Messages API (tool use)
Shared — packages/shared
zod 4 schemas · normalisation · validation

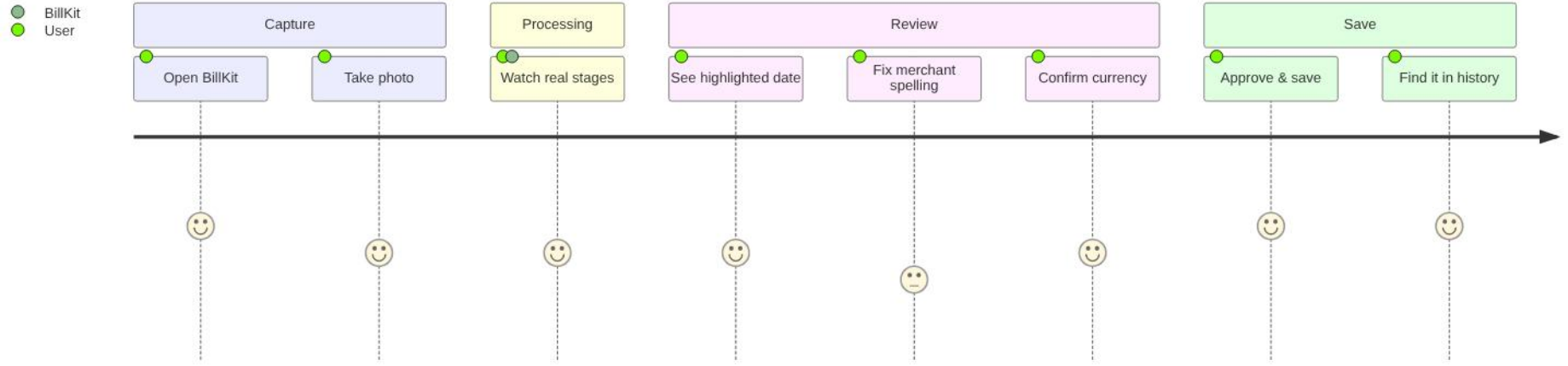

Vite + React 19 SPA
Service worker
/app/*
authenticated
pages
lazy-loaded
chunks
Public pages
prerendered
HTML for SEO
services: api ·
auth · toast ·
offline · theme
fetch /api, same origin
Fastify backend

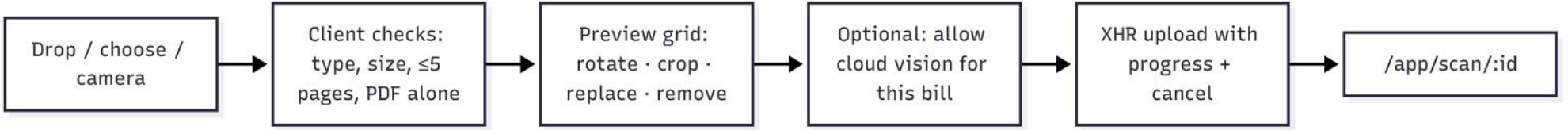

main.tsx
AuthProvider
ToastProvider
AppRoutes
MarketingLayout
Auth pages: Login · Register · Forgot · Reset
AppLayout: sidebar · topbar search · bottom nav · offline banner
Home · Features · How it works · Technology · Privacy · About · FAQ · Contact · Legal
Dashboard · Scan · ScanDetail(Processing/Rejected/Review) · Bills · BillDetail · Analytics · Categories · Settings · Admin
Components: ReviewBadge · Dialog · Form fields · Charts · Empty states · Camera/Crop dialogs

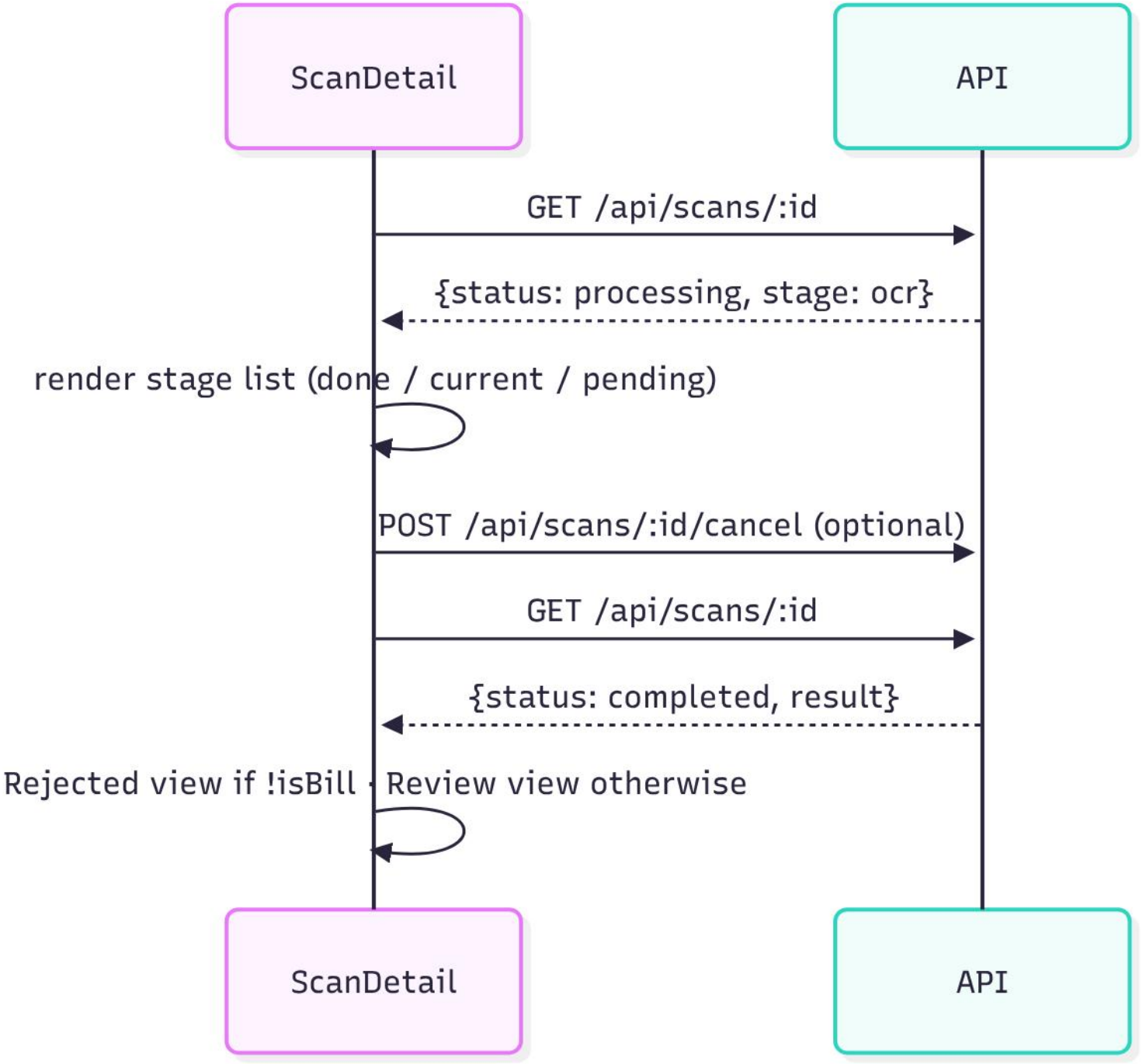

/
/features
/how-it-works
/technology
/privacy-security
/about
/faq
/contact
/legal/*
/register
/login
/app
/app/scan
/app/scan/:id
/app/bills/:id
/app/bills
/app/analytics
/app/categories
/app/settings
role=admin
/app/admin
/forgot-password
/reset-password

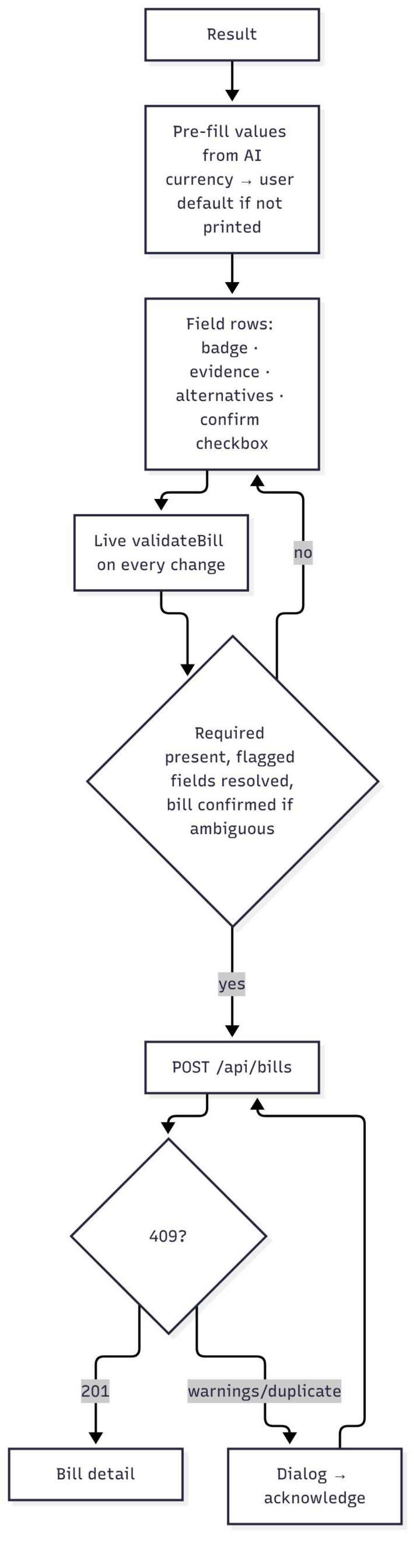

Anonymous
/register
/login
/forgot-password
token used, sessions revoked
Registering
401 generic error
SigningIn
Resetting
Sign out (cookie cleared, IndexedDB cleared)
Session expired → redirect /login?next=...
201 + session cookie
200
SignedIn
Delete account

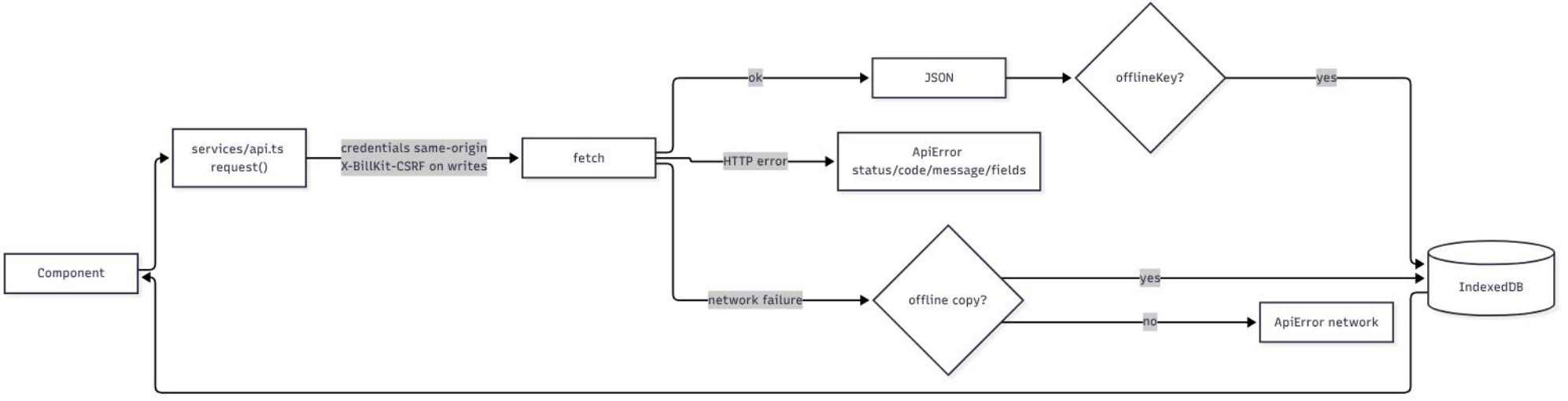
Scanning a restaurant bill

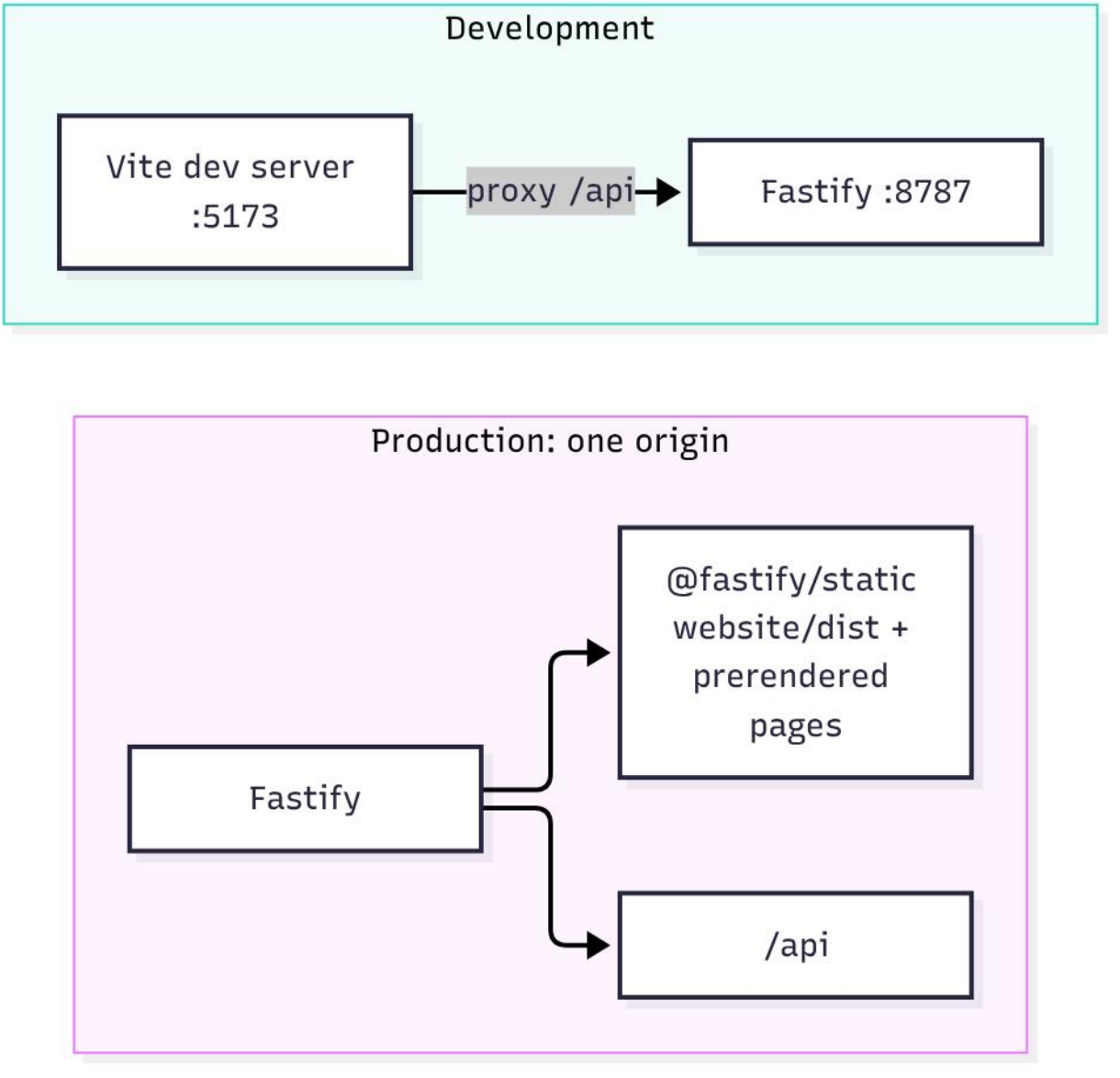


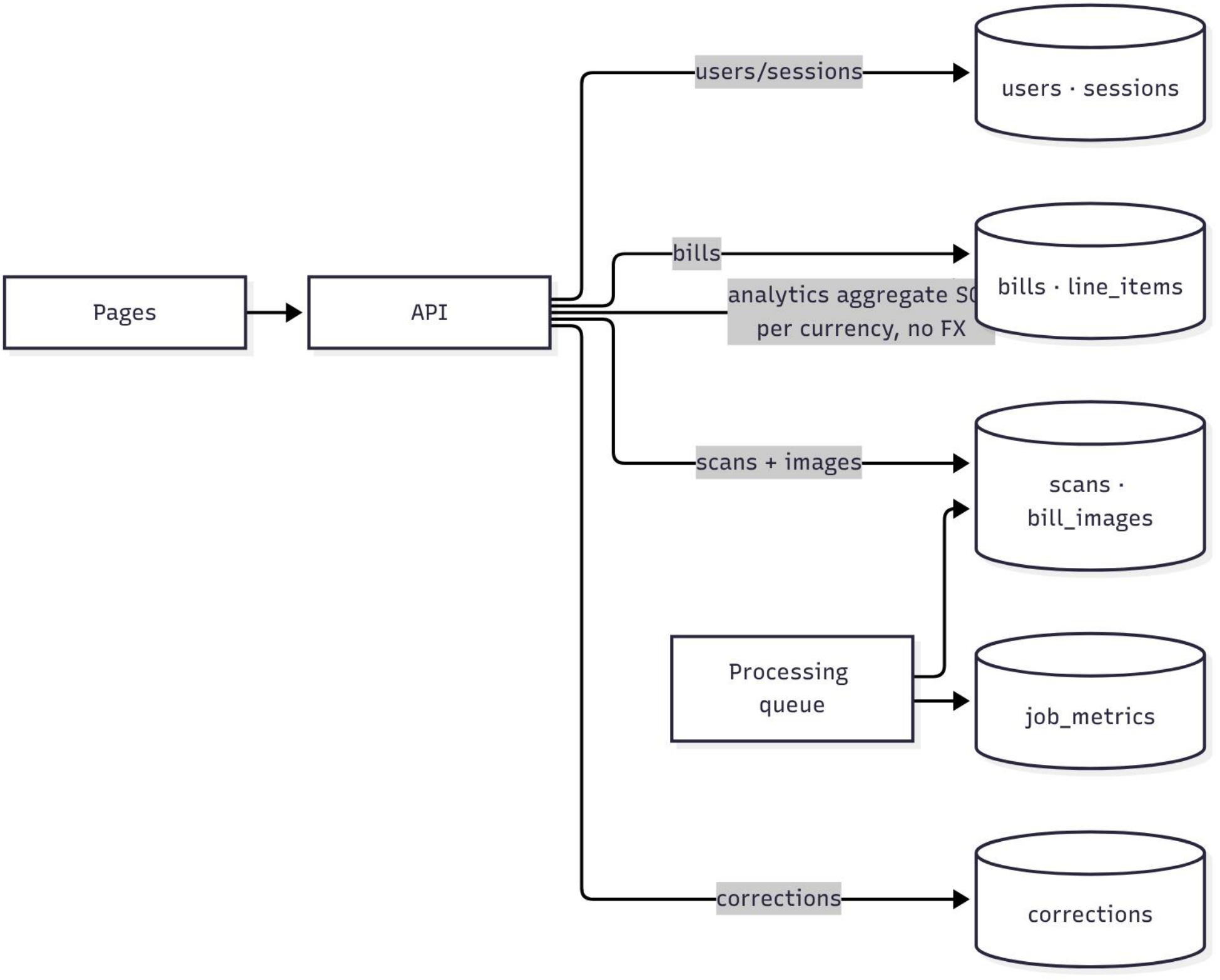

Drop / choose / camera
Client checks: type, size, ≤5 pages, PDF alone
Preview grid: rotate · crop · replace · remove
Optional: allow cloud vision for this bill
XHR upload with progress + cancel
/app/scan/:id

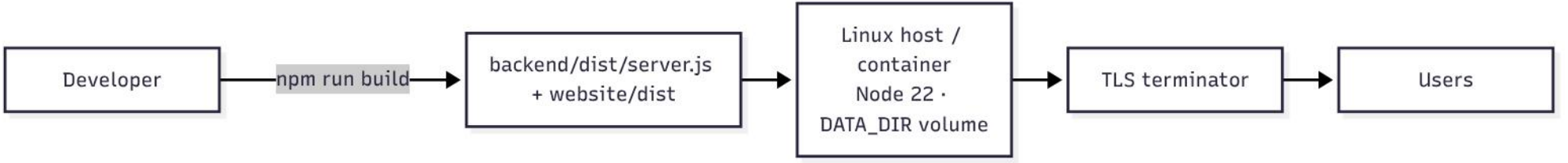

ScanDetail
API
GET /api/scans/:id
{status: processing, stage: ocr}
render stage list (done / current / pending)
POST /api/scans/:id/cancel (optional)
GET /api/scans/:id
{status: completed, result}
Rejected view if !isBill · Review view otherwise
ScanDetail
API

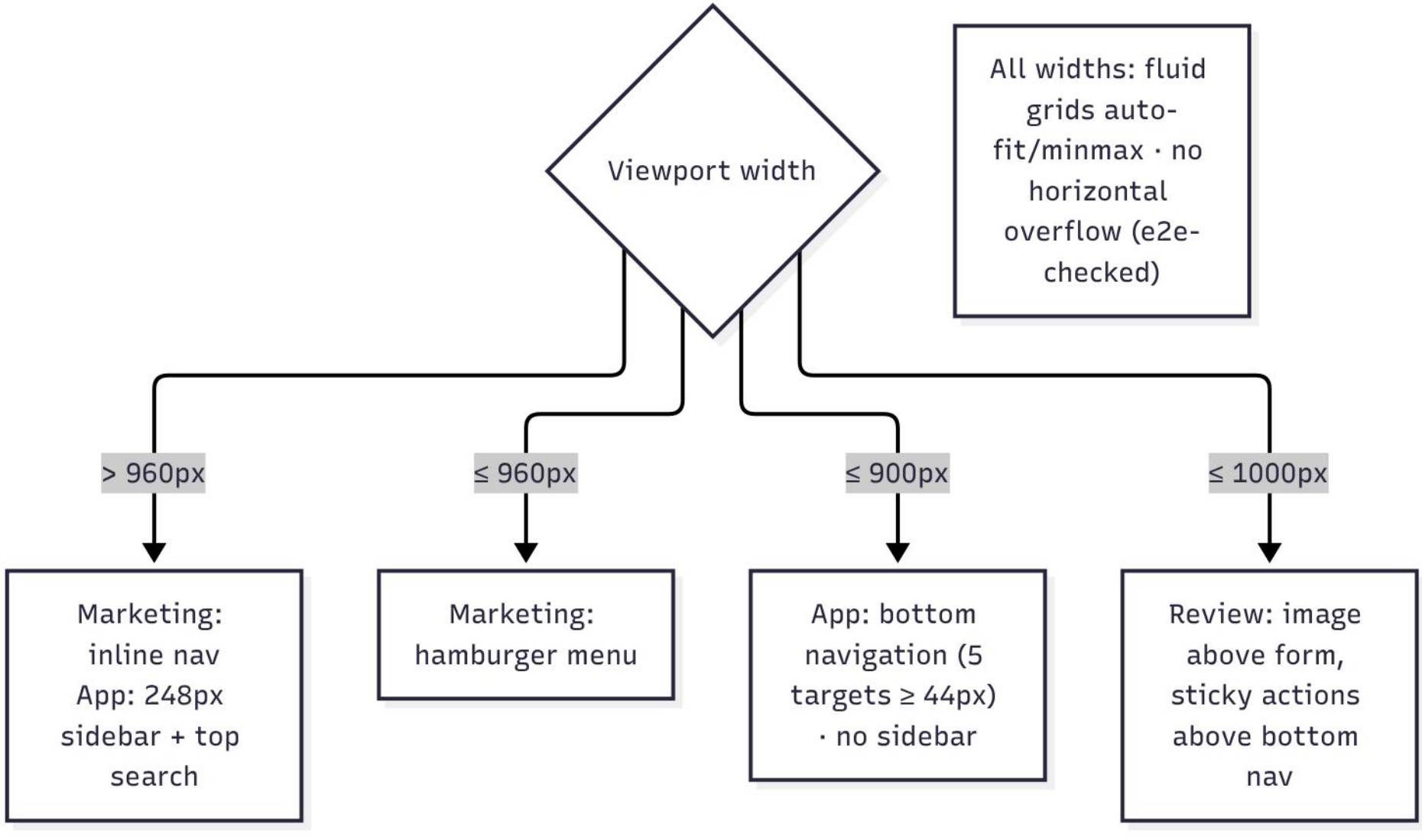

Result
Pre-fill values from AI currency → user default if not printed
Field rows: badge · evidence · alternatives · confirm checkbox
Live validateBill on every change
no
Required present, flagged fields resolved, bill confirmed if ambiguous
yes
POST /api/bills
409?
201
warnings/duplicate
Bill detail
Dialog → acknowledge

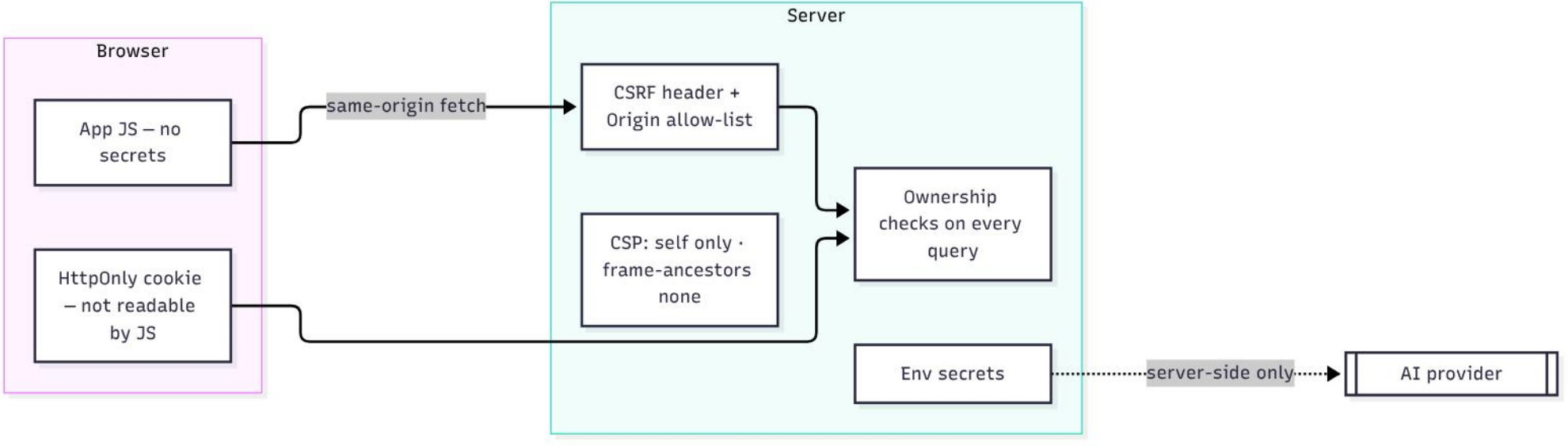

Component
services/api.ts
request()
credentials same-origin
X-BillKit-CSRF on writes
fetch
ok
JSON
offlineKey?
yes
HTTP error
ApiError
status/code/message/fields
network failure
offline copy?
yes
no
ApiError network
IndexedDB


Development
Vite dev server :5173
proxy /api
Fastify :8787
Production: one origin
Fastify
@fastify/static website/dist + prerendered pages
/api


Pages
API
users/sessions
users · sessions
bills
bills · line_items
analytics aggregate SQ
per currency, no FX
scans + images
scans ·
bill_images
Processing
queue
job_metrics
corrections
corrections


Developer
npm run build
backend/dist/server.js
+ website/dist
Linux host /
container
Node 22 ·
DATA_DIR volume
TLS terminator
Users


Viewport width
All widths: fluid grids auto-fit/minmax · no horizontal overflow (e2e-checked)
> 960px
≤ 960px
≤ 900px
≤ 1000px
Marketing: inline nav App: 248px sidebar + top search
Marketing: hamburger menu
App: bottom navigation (5 targets ≥ 44px) · no sidebar
Review: image above form, sticky actions above bottom nav


Browser
App JS — no secrets
HttpOnly cookie — not readable by JS
same-origin fetch
Server
CSRF header + Origin allow-list
CSP: self only · frame-ancestors none
Ownership checks on every query
Env secrets
server-side only
AI provider